

# Boundary currents and compact quantum homotopies

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# Holomorphic BF theory and its boundary current

A current algebra can lose its central term while its bulk field theory retains a nondegenerate Batalin–Vilkovisky pairing. The mechanism is a change in the differential, with two field components kept separate. On a complex curve times a real half-line, the quadratic action density is

$$\mathcal{L}_\kappa(a, b) = b \wedge d' a + \frac{\kappa}{2} a \wedge \partial a, \quad d' = \bar{\partial} + d_t.$$

It is defined for smooth fields. On compactly supported fields the action is  $S_\kappa(a, b) = \int_{\mathcal{M}}^{\text{res}} \mathcal{L}_\kappa(a, b)$ . Here  $a$  is a partial connection in the antiholomorphic and real directions, and  $b$  is a holomorphic one-form. The pairing between these fields is independent of  $\kappa$ . At  $\kappa = 0$  the action is holomorphic BF theory: the field  $b$  multiplies the partial curvature  $d' a$ . A compactly supported collar map identifies its quantum boundary observables with the boundary value of the bulk theory. A Cauchy contraction then extracts every current field and operator product from those supported observables. With the positive unary differential fixed below, the boundary current has level  $-\kappa$ . At zero its singular product vanishes, while its polynomial fields and all holomorphic jets remain.

## 1.1 THE FIELDS, INCLUDING THEIR ODD COMPONENTS

Let  $\Sigma = \mathbb{C}$  with coordinate  $z = x + iy$ , and let  $\mathcal{M} = \Sigma \times \mathbb{R}_{\geq 0}$  with coordinate  $t$  on the second factor. Choose the orientation whose induced boundary orientation on  $\Sigma \times \{0\}$  is the complex orientation. Write  $\mathcal{A}$  for smooth forms generated by  $d\bar{z}, dt$ . All coefficients are complex-valued, and

$$d' = d\bar{z} \wedge \partial_{\bar{z}} + dt \wedge \partial_t, \quad \partial = dz \wedge \partial_z.$$

These operators satisfy  $(d')^2 = \partial^2 = d' \partial + \partial d' = 0$ . The parameter  $\kappa \in \mathbb{C}$  is a scalar of cohomological degree zero. Statements polynomial in  $\kappa$  also hold over  $\mathbb{C}[\kappa]$ .

The coefficient space of  $\alpha$  is  $\mathbb{C}e$ , and that of  $\beta$  is its dual  $\mathbb{C}e^\vee$ . These chosen bases suppress the evaluation pairing in the scalar formulas. The fields have two summands,

$$E_\kappa = \mathcal{A}[1] \oplus (\mathcal{A} \wedge dz)[1].$$

This is the unrestricted smooth field complex. Write  $E_{\kappa, c}$  for its subcomplex of compactly supported fields; compact support is required in both the curve and the real direction and may meet the boundary

of  $M$ . The notation records the following precise degrees: an ordinary  $p$ -form in either summand has cohomological degree  $p - 1$ . The differential is the actual exterior operator

$$Q_\kappa(\alpha, \beta) = (d'\alpha, d'\beta + \kappa \partial \alpha).$$

Thus the holomorphic one-form  $\beta$  has degree zero before adding any  $d\bar{z}$  or  $dt$ . All tensor substitutions use the Koszul sign of the displayed cohomological degrees.

For a component of the field complex, the corresponding coordinate has the opposite degree. This coordinate degree is the ghost number. The complete list is

| component          | form degree | ghost number | role                 |
|--------------------|-------------|--------------|----------------------|
| $\alpha^{(0)}$     | 0           | 1            | gauge ghost $c$      |
| $\alpha^{(1)} = a$ | 1           | 0            | partial connection   |
| $\beta^{(1)} = b$  | 1           | 0            | holomorphic one-form |
| $\alpha^{(2)}$     | 2           | -1           | antifield of $b$     |
| $\beta^{(2)}$      | 2           | -1           | antifields of $a$    |
| $\beta^{(3)}$      | 3           | -2           | antifield of $c$ .   |

There is no  $\alpha^{(3)}$ , because  $\mathcal{A}$  has only two exterior generators. Both summands are essential to the BF presentation.

To fix the current normalization, define

$$\int^{\text{res}} = \frac{1}{2\pi i} \int$$

on compactly supported complex-valued top forms, both on  $M$  and on  $\Sigma$ . The second integral is ordinary oriented integration. In contour notation  $\text{Res}_{z=0} z^{-1} dz = 1$ . For a compactly supported two-form  $\theta$  on  $\Sigma$  and a compactly supported function  $\rho$  on the half-line, fiber integration has the sign

$$\int_M^{\text{res}} \theta \wedge \rho(t) dt = - \left( \int_0^\infty \rho(t) dt \right) \int_\Sigma^{\text{res}} \theta.$$

Indeed, the orientation of  $M$  is the negative of the product orientation on  $\Sigma \times \mathbb{R}_{\geq 0}$ . Changing the common scalar normalization rescales the action and inverse pairing together.

For homogeneous smooth fields  $u = (\alpha, \beta)$  of form degree  $p$  and  $v = (\alpha', \beta')$ , define the local pairing density

$$\omega_{\text{loc}}(u, v) = (-1)^{p+1} (\alpha \wedge \beta' + \beta \wedge \alpha')_{[3]},$$

where  $[3]$  takes the top-form component. The orientation identifies this top form with a complex-valued density. Its integral

$$\omega(u, v) = \int_M^{\text{res}} \omega_{\text{loc}}(u, v)$$

is defined whenever at least one argument has compact support. Thus the integrated pairing is defined on  $E_{\kappa, c} \otimes E_\kappa$  and on  $E_\kappa \otimes E_{\kappa, c}$ ; no integral is assigned to an arbitrary pair of unrestricted smooth fields. The local pairing has degree  $-1$ , is independent of  $\kappa$ , and is fiberwise nondegenerate: the two complementary summands pair by evaluation and wedge product. Its sign gives graded skew-symmetry. Indeed, a nonzero pairing has  $p + q = 3$ , so interchanging the forms leaves their wedge sign unchanged and changes  $(-1)^{p+1}$  to its negative.

The distinction is already necessary for fields away from the boundary. If  $\chi$  is a nonzero real smooth function supported in  $(1, 2)$ , then  $u = (\chi(t)dt, 0)$  and  $v = (0, \chi(t)dz \wedge d\bar{z})$  have a well-defined local pairing density, but its integral contains the infinite area of  $\mathbb{C}$ . Their support is not compact in  $M$ .

## 1.2 THE BV IDENTITY AND THE CHIRAL BOUNDARY CONDITION

The square of the differential is zero for every  $\kappa$ :

$$Q_\kappa^2(\alpha, \beta) = (\circ, \kappa(d'\partial + \partial d')\alpha) = \circ.$$

The linear complex is elliptic, including at zero level. For a nonzero real covector  $\xi$ , the symbol of  $d'$  is wedge multiplication by  $\xi^{\circ,1} + \xi_t dt$ . This one-form is nonzero: a real covector with zero  $(\circ, 1)$  part and zero  $t$  part is zero. Contraction with a dual vector contracts this symbol complex. Choose that vector with zero  $z$  component. Its contraction anticommutes with the symbol of  $\partial$ , so it also contracts the symbol of  $Q_\kappa$ .

**Proposition 1.1** (A nondegenerate BV family). *For every  $\kappa \in \mathbb{C}$ , the fields, differential and local pairing above define a free BV theory in the interior of  $M$ . The boundary condition*

$$L = \Omega^{\bullet}(\Sigma), \quad \iota^* \alpha = \circ, \quad \iota^* \beta \in L, \quad \iota : \Sigma \hookrightarrow M,$$

*is a local Lagrangian boundary condition. The restricted differential is compatible with the integrated pairing whenever at least one argument has compact support.*

*Proof.* Ellipticity and nondegeneracy were proved above. For homogeneous  $u$  of form degree  $p$ , the exterior Leibniz rule gives the local identity

$$\begin{aligned} \omega_{\text{loc}}(Q_\kappa u, v) + (-1)^{p-1} \omega_{\text{loc}}(u, Q_\kappa v) \\ = (-1)^p [d'(\alpha \wedge \beta' + \beta \wedge \alpha') + \kappa \partial(\alpha \wedge \alpha')]_{[3]}. \end{aligned}$$

Assume now that at least one of  $u, v$  has compact support. The primitives on the right then have compact support, so their Stokes integrals have no contribution from infinity. The second term integrates to zero: its possible primitive is a two-form generated by  $d\bar{z}, dt$ , whose boundary pullback vanishes. The first term is an ordinary total derivative, since applying  $\partial$  to its primitive gives holomorphic degree two. Consequently

$$\begin{aligned} \omega(Q_\kappa u, v) + (-1)^{p-1} \omega(u, Q_\kappa v) \\ = (-1)^p \int_\Sigma^{\text{res}} \iota^*(\alpha \wedge \beta' + \beta \wedge \alpha'). \end{aligned}$$

Within this support domain the expression vanishes if one argument is supported away from the boundary, or if both satisfy  $L$ .

On the boundary, the two Dolbeault summands have equal rank and pair nondegenerately with one another. The second summand is isotropic, so it is Lagrangian. It is preserved by  $Q_\kappa$  because  $\partial\beta = \circ$ . Its differential is the Dolbeault operator on holomorphic one-forms, an elliptic complex on  $\Sigma$ . The same decomposition tensored with normal forms makes the theory topological in the normal direction. When shifts precede this tensor product, the identification of a normal one-form carries the usual suspension sign; equivalently one may use the exterior-form formula for  $Q_\kappa$  displayed above. Thus the boundary condition is a Lagrangian elliptic subcomplex, and the bulk complex is topological in its normal direction.  $\square$

Fix the Hamiltonian convention  $\iota_{Q_\kappa} \omega = -\delta S$ . The full graded smooth field complex has the local quadratic density

$$S_{\text{loc}}(\Phi) = \frac{1}{2} \omega_{\text{loc}}(\Phi, Q_\kappa \Phi).$$

For  $\Phi \in E_{\kappa, c}$  its integral is the well-defined functional  $S(\Phi) = \int_M^{\text{res}} S_{\text{loc}}(\Phi) = \frac{1}{2} \omega(\Phi, Q_\kappa \Phi)$ . Write  $E_{\kappa, L, c} = \{\Phi \in E_{\kappa, c} : \iota^* \alpha = \circ\}$ . With the chiral boundary condition imposed, this functional and its

Hamiltonian identity are taken on that space. For unrestricted fields, the action is interpreted as a local density, whose first variation is tested against compactly supported variations.

On ghost-number-zero components the density is

$$\frac{1}{2}(a \wedge d'b + b \wedge d'a + \kappa a \wedge \partial a) = \mathcal{L}_\kappa(a, b) - \frac{1}{2}d'(a \wedge b).$$

For compactly supported fields satisfying  $i^*a = 0$ , Stokes' theorem kills the total derivative, so its integral is the action stated at the start of the chapter. The local Euler–Lagrange equations, valid on the unrestricted smooth field complex, are

$$d'a = 0, \quad d'b + \kappa \partial a = 0.$$

Their local gauge symmetry, with smooth gauge parameter  $c$  vanishing on the boundary, is

$$\delta_c a = d'c, \quad \delta_c b = \kappa \partial c.$$

A compactly supported  $c$  also preserves the compactly supported field space. The second equation is invariant because  $d'\partial = -\partial d'$ .

For any smooth configuration satisfying  $L$  and any compactly supported variation  $(\delta a, \delta b)$  satisfying the same boundary condition, the first variation of the local action integrates to

$$\begin{aligned} \int_M^{\text{res}} \delta \mathcal{L}_\kappa &= \int_M^{\text{res}} (\delta b \wedge d'a + \delta a \wedge (d'b + \kappa \partial a)) \\ &\quad - \int_\Sigma^{\text{res}} b \wedge \delta a. \end{aligned}$$

All integrands have compact support because each contains a variation or its derivative. Integration by parts is therefore legitimate, and the boundary term vanishes. This proves the variational Hamiltonian identity with its support domain. The configuration  $a = t d\bar{z}$ ,  $b = dz$  illustrates the remaining distinction: it satisfies the boundary condition but has a nonintegrable constant action density. Its local variational equations are defined even though it has no integrated action value.

The terms involving ghosts and antifields are fixed by  $S_{\text{loc}}$  on the unrestricted graded fields. On  $E_{\kappa, L, c}$  the integrated functional satisfies the classical master equation: the Hamiltonian vector field of  $\{S, S\}$  is  $2Q_\kappa^2 = 0$ , and this quadratic functional has no constant term. Equivalently, the same statement is an identity of local variational functionals when evaluated against compactly supported variations.

### 1.3 THE NONZERO-LEVEL CHANGE OF FIELDS AND ITS SPECIAL FIBER

For  $\kappa \neq 0$ , put

$$F_\kappa(\alpha, \beta) = \alpha + \kappa^{-1}\beta.$$

The target is the full de Rham field complex  $\Omega^\bullet(\mathcal{M})[1]$  with differential  $d = d' + \partial$  and local pairing density  $\omega_{\text{CS}, \kappa, \text{loc}}(u, v) = (-1)^{p+1} \kappa(u \wedge v)_{[3]}$ . Its integrated pairing is  $\omega_{\text{CS}, \kappa} = \int_M^{\text{res}} \omega_{\text{CS}, \kappa, \text{loc}}$ , with at least one compactly supported argument.

**Proposition 1.2** (Comparison with abelian Chern–Simons theory). *For every nonzero complex  $\kappa$ ,  $F_\kappa$  is an isomorphism of linear BV theories with chiral boundary condition. It identifies the local BV Hamiltonian densities and the integrated BF and level- $\kappa$  abelian Chern–Simons actions on compactly supported fields satisfying that boundary condition. This field isomorphism has no specialization at  $\kappa = 0$ .*

*Proof.* The inverse separates the holomorphic degrees and multiplies the holomorphic-one-form summand by  $\kappa$ . Furthermore,

$$dF_\kappa(\alpha, \beta) = d'\alpha + \partial\alpha + \kappa^{-1}d'\beta = F_\kappa Q_\kappa(\alpha, \beta).$$

The product of two  $\alpha$  summands has no three-form component. The product of two  $\beta$  summands has holomorphic degree two and vanishes. The two cross terms consequently give  $F_\kappa^* \omega_{\text{CS}, \kappa, \text{loc}} = \omega_{\text{loc}}$  pointwise. Both  $F_\kappa$  and its inverse preserve compact support, so the same identity holds for the integrated pairings on their stated domains. The local Hamiltonian densities, the integrated actions on compactly supported fields, and the boundary condition are preserved. The inverse scalar  $\kappa^{-1}$  in  $F_\kappa$  excludes the zero fiber.  $\square$

At zero level the field pairing remains the same nondegenerate cross pairing. In particular,  $b$  remains a field, rather than becoming  $\kappa$  times a fixed field. Its gauge transformation vanishes, while the gauge transformation of  $a$  remains  $d'c$ .

This difference can be measured before quantization. On a product of a complex disk  $D$  and an interior real interval, the local cohomology of the field complex is computed by

$$O(D) \xrightarrow{\kappa\partial} \Omega_{\text{hol}}^1(D),$$

placed in degrees  $-1, 0$ . The Dolbeault and real Poincaré homotopies remove the antiholomorphic and real directions. At nonzero  $\kappa$ , holomorphic integration on  $D$  makes the arrow surjective and its kernel consists of constants. At zero it is the zero map. Thus

|                 |                    |                              |
|-----------------|--------------------|------------------------------|
|                 | $H^{-1}(E_\kappa)$ | $H^0(E_\kappa)$              |
| $\kappa \neq 0$ | $\mathbb{C}$       | $0$                          |
| $\kappa = 0$    | $O(D)$             | $\Omega_{\text{hol}}^1(D)$ . |

These are cohomology groups of the field complex. The negative-degree summand records gauge data. The zero fiber therefore has more local field cohomology than the nonzero Chern–Simons fibers.

#### 1.4 A GREEN OPERATOR AT ZERO LEVEL

The zero-level kinetic complex has a concrete propagator operator in the interior. Give  $\mathbb{C} \times \mathbb{R}$  its flat Euclidean metric and put

$$Q^{\text{GF}} = -4\partial_{\bar{z}}\iota_{d\bar{z}} - \partial_t\iota_{dt}.$$

The contraction symbols satisfy  $\iota_{d\bar{z}}(d\bar{z}) = \iota_{dt}(dt) = 1$  and kill the other basis one-forms. Then

$$Q_0 Q^{\text{GF}} + Q^{\text{GF}} Q_0 = -4\partial_z\partial_{\bar{z}} - \partial_t^2 = -\partial_x^2 - \partial_y^2 - \partial_t^2 =: \Delta.$$

The scalar Green kernel is  $G(X, Y) = (4\pi|X - Y|)^{-1}$ , with  $\Delta_X G = \delta_Y$ . Consequently  $H_0 = Q^{\text{GF}} G$  satisfies

$$Q_0 H_0 + H_0 Q_0 = 1$$

on compactly supported test forms, with output interpreted as smooth forms with the corresponding decay. Its kernel is

$$H_0(X, Y) = \frac{\overline{z - w}}{2\pi|X - Y|^3} \iota_{d\bar{z}} + \frac{t - s}{4\pi|X - Y|^3} \iota_{dt}, \quad X = (z, t), \quad Y = (w, s).$$

The distribution identity follows from the outward flux of  $-\nabla(4\pi r)^{-1}$  through a sphere, which is one. Differentiating  $G$  gives the two displayed coefficients. No inverse power of  $\kappa$  occurs. Composing with the

inverse cross pairing gives the BV contraction kernel, including the constant  $2\pi i$  from the normalization of  $\omega$ .

This interior operator is not a claim about a boundary Green problem. The boundary comparison below uses compactly supported normal homotopies which preserve the chiral boundary condition. Those homotopies require neither a choice of boundary Green kernel nor a limit of  $\kappa^{-1}$ .

### 1.5 THE BOUNDARY COMPLEX AND ITS QUANTUM COCYCLE

Write  $\mathbf{j}(U) = \Omega_c^{\circ, \bullet}(U)$  for an open  $U \subseteq \Sigma$ . A generator  $u \in \Omega_c^{\circ, q}(U)$  has degree  $q - 1$  in the boundary observable complex. The boundary complement is the antiholomorphic summand, while the component of the boundary differential from this complement to  $L$  is  $\kappa \partial$ . Let  $\hbar$  be a central polynomial parameter of cohomological degree zero. The boundary observables are

$$\mathcal{B}_\kappa(U) = \left( \text{Sym}(\mathbf{j}(U)[1])[\hbar], \bar{\partial} + \hbar \Delta_{\mu_\kappa} \right), \quad \mu_\kappa(u, v) = -\kappa \int_U^{\text{res}} u \wedge \partial v.$$

The pairing is zero unless the antiholomorphic degrees sum to one. As a contraction of observable generators, it has degree one. The operator  $\Delta_{\mu_\kappa}$  vanishes on constants and linear terms; on a monomial it sums contractions of unordered pairs, with the Koszul sign for bringing that pair together. This specifies all its values. The differential on a shifted generator is the displayed  $\bar{\partial}$ , transported without an additional minus sign.

The completion in each symmetric power has a concrete meaning. For the  $r$ th tensor power, take compactly supported smooth sections on  $U^r$  of the external tensor product of the field bundles; then pass to coinvariants for the graded action of the symmetric group. On every fixed compact support use the seminorms of all derivatives, and take the direct limit over compact supports. This is the smooth kernel, or completed bornological tensor, convention. The symmetric algebra is the direct sum of these powers. Thus polynomial degree is finite; there is no completion in the number of fields.

**Lemma 1.3** (Boundary quantum differential). *The square of  $\bar{\partial} + \hbar \Delta_{\mu_\kappa}$  vanishes for every  $\kappa$ . At  $\kappa = 0$ , the quantum boundary complex is exactly the commutative Dolbeault algebra  $\text{Sym}(\mathbf{j}(U)[1])[\hbar]$ .*

*Proof.* For  $u$  of antiholomorphic degree  $q$ , the compatibility equation for one contraction is

$$\mu_\kappa(\bar{\partial} u, v) + (-1)^{q-1} \mu_\kappa(u, \bar{\partial} v) = -\kappa \int_U^{\text{res}} \bar{\partial}(u \wedge \partial v) = 0.$$

Compact support removes the boundary integral. It follows that  $\bar{\partial} \Delta_{\mu_\kappa} + \Delta_{\mu_\kappa} \bar{\partial} = 0$  on every symmetric power. In the square of  $\Delta_{\mu_\kappa}$ , overlapping pairs give zero because a contracted factor has disappeared. Disjoint pairs occur in both orders with opposite signs, since a contraction has odd degree. Thus  $\Delta_{\mu_\kappa}^2 = 0$ . Finally,  $\mu_0 = 0$  by its formula.  $\square$

The coefficient  $\kappa$  comes from the differential  $\kappa \partial$ , not from a degeneration of the field pairing. The next calculation derives this cocycle directly from bulk fields.

### 1.6 A COMPACT COLLAR MAP FOR QUANTUM OBSERVABLES

Fix a smooth compactly supported function  $\rho$  on  $\mathbb{R}_{\geq 0}$  with integral one, and put  $R(t) = \int_0^t \rho(s) ds$ . Thus  $R = 1$  for sufficiently large  $t$ . Let  $E_{\kappa, L, c}(U \times \mathbb{R}_{\geq 0})$  denote compactly supported fields satisfying  $\iota^* \alpha = 0$ . For  $u \in \Omega_c^{\circ, q}(U)$  define

$$I_\kappa(u) = u \wedge \rho(t) dt + (-1)^q (R(t) - 1) \kappa \partial u.$$

This is a degree-zero map from the unshifted complex  $(\mathfrak{j}(U), \bar{\partial})$  to  $(E_{\kappa, L, c}(U \times \mathbb{R}_{\geq 0}), Q_{\kappa})$ . The first term is in the  $\alpha$  summand and the second is in the  $\beta$  summand. Both have compact support. The first has zero boundary pullback, and the second lies in  $L$ .

The bulk observable generators are  $sx$ , for  $x \in E_{\kappa, L, c}$ , with  $|sx| = |x| - 1$ . Define their contraction by

$$B(sx, sy) = (-1)^{|x|} \omega(x, y).$$

For complementary form degrees this is the normalized integral of the two cross terms. Let  $\Delta_B$  sum pair contractions as above. The bulk–boundary quantum complex is

$$\mathcal{O}_{\kappa}(U \times \mathbb{R}_{\geq 0}) = (\text{Sym}(E_{\kappa, L, c}[1])[\hbar], Q_{\kappa} + \hbar \Delta_B).$$

The shifted unary operator is  $Q_{\kappa}(sx) = sQ_{\kappa}x$ . The pairing identity in Proposition 1.1 and the same disjoint-pair cancellation show that this differential squares to zero.

**Theorem 1.4** (The quantum boundary map at every level). *The maps  $I_{\kappa}$  induce quasi-isomorphisms*

$$\text{Sym}(I_{\kappa}) : \mathcal{B}_{\kappa}(U) \longrightarrow \mathcal{O}_{\kappa}(U \times \mathbb{R}_{\geq 0})$$

*for every open  $U \subseteq \Sigma$  and every complex  $\kappa$ , including zero. They commute with extension by zero and with factorization products on disjoint opens.*

*Proof.* First check the unary differential. Differentiating the first term of  $I_{\kappa}(u)$  gives  $\bar{\partial}u \wedge \rho dt + \kappa \partial u \wedge \rho dt$ . Differentiating the second gives

$$-(-1)^q (R - 1) \kappa \partial \bar{\partial} u - \kappa \partial u \wedge \rho dt.$$

The last terms cancel, leaving  $I_{\kappa}(\bar{\partial}u)$ .

Projection to the  $\alpha$  summand followed by integration along the normal one-form gives a cochain map  $P$ , with  $PI_{\kappa} = 1$ . To prove that  $P$  is a quasi-isomorphism, consider the exact sequence whose subcomplex is the  $\beta$  summand and whose quotient is the  $\alpha$  summand. The former has absolute normal forms; the latter has relative normal forms, whose functions vanish at  $t = 0$ . On the normal complexes the homotopies are

$$\begin{aligned} h_{\text{abs}}(g dt) &= - \int_t^{\infty} g(s) ds, \\ h_{\text{rel}}(g dt) &= \int_0^t g(s) ds - R(t) \int_0^{\infty} g(s) ds. \end{aligned}$$

Both vanish on functions. The absolute homotopy satisfies  $dh + hd = 1$ . The relative one satisfies  $dh + hd = 1 - ip$ , where  $p(g dt) = \int g$  and  $i(1) = \rho dt$  in degree one. Their values are compactly supported, and the relative value vanishes at  $t = 0$ . Direct differentiation proves all these identities. For a coefficient of tangential form degree  $r$ , multiply the normal homotopy by  $(-1)^r$ ; this cancels the tangential cross terms. These formulas also apply to smooth sections depending on  $z$ . The  $\beta$  subcomplex is therefore acyclic, and the relative  $\alpha$  quotient retracts onto  $\mathfrak{j}(U)$  after its displayed shift. The long exact sequence proves that  $P$ , hence  $I_{\kappa}$ , is a quasi-isomorphism.

There is also a continuous contraction for the full split complex. Let  $H$  be the two normal homotopies just displayed, and put  $\partial_{\kappa} = Q_{\kappa} - Q_0$ . The correction goes from the  $\alpha$  summand to the  $\beta$  summand, so  $\partial_{\kappa} H \partial_{\kappa} = 0$  and  $P \partial_{\kappa} = 0$ . Consequently

$$H_{\kappa} = H - H \partial_{\kappa} H, \quad I_{\kappa} = I_0 - H \partial_{\kappa} I_0, \quad Q_{\kappa} H_{\kappa} + H_{\kappa} Q_{\kappa} = 1 - I_{\kappa} P.$$

To check the last identity, substitute  $Q_0 H + H Q_0 = 1 - I_0 P$  and  $Q_0 \partial_{\kappa} = -\partial_{\kappa} Q_0$ . The linear correction terms cancel except for  $H \partial_{\kappa} I_0 P$ ; the quadratic correction terms vanish. The formula for  $I_{\kappa}$  is the displayed collar map, since the absolute homotopy of  $\rho dt$  is  $R - 1$ . This construction uses no division by  $\kappa$ .

It remains to check the quantum contraction, rather than only the unary differential. For antiholomorphic degrees  $q + r = 1$ , expand the two cross terms in  $I_\kappa(u) \wedge I_\kappa(v)$ . Moving  $dt$  to the right, and integrating by parts on  $U$ , gives

$$B(sI_\kappa(u), sI_\kappa(v)) = -2\kappa \left( \int_0^\infty \rho(t)(1 - R(t))dt \right) \int_U^{\text{res}} u \wedge \partial v.$$

The minus sign is the fiber-integration sign fixed above;  $B$  itself integrates the two cross terms with positive sign. For the second term use  $\int_U \partial u \wedge v = (-1)^{q+1} \int_U u \wedge \partial v$ . The normal integral is

$$\int_0^\infty \rho(1 - R)dt = \left[ R - \frac{R^2}{2} \right]_0^\infty = \frac{1}{2}.$$

Consequently  $B(I_\kappa(u), I_\kappa(v)) = \mu_\kappa(u, v)$ , with suspensions understood. This proves compatibility on quadratic monomials. The pair-contraction definitions prove it on every symmetric power.

At the classical level, the preceding continuous linear homotopies extend to  $r$  tensor factors by the explicit map

$$\mathcal{H}_r = \sum_{j=1}^r (I_\kappa P)^{\otimes(j-1)} \otimes H_\kappa \otimes 1^{\otimes(r-j)}.$$

Use the Koszul sign when the degree-minus-one map  $H_\kappa$  passes the preceding factors. In  $d\mathcal{H}_r + \mathcal{H}_r d$ , the terms  $1 - I_\kappa P$  telescope, giving  $1 - (I_\kappa P)^{\otimes r}$ . Averaging over the finite symmetric group gives the result on symmetric powers; characteristic zero permits this averaging. The integrations preserve compact supports in the product variables and extend to the completed bornological tensors. Thus  $\text{Sym}(I_\kappa)$  is a classical quasi-isomorphism.

Finally filter both polynomial quantum complexes by the sum of the power of  $\hbar$  and the number of fields. The unary differential preserves that filtration degree. A quantum contraction reduces it by one. The associated graded map is the classical quasi-isomorphism tensored with  $\mathbb{C}[\hbar]$ . The filtration starts in degree zero, and every element has finite filtration degree. Induction on the finite filtration, followed by the exact direct limit, proves the quantum quasi-isomorphism. All formulas use the same  $\rho$  and integrate only the normal variable. They therefore commute with extension by zero and with products on disjoint boundary opens.  $\square$

The observables also satisfy the gluing property needed for a factorization algebra. The relevant covers  $\{U_i\}$  of  $U$  are those for which every finite subset of  $U$  lies in some  $U_i$ . For every  $r$ , the sets  $U_i^r$  then cover  $U^r$ . At a fixed field degree the augmented Čech complex of compactly supported smooth kernels for this cover is exact: a locally finite partition of unity gives its contraction, and only finitely many terms meet a given compact support. Multiplication by the partition functions preserves the linear boundary condition for bulk fields. For fixed  $r$  the form degrees are bounded, so the finite filtration by form degree upgrades this Čech exactness to descent for the unary differential. Passing to symmetric coinvariants preserves exactness because averaging over the finite symmetric group is available. The polynomial filtration used above then proves quantum descent. Here factorization algebra means precisely these gluing complexes together with the compatible products for disjoint opens. Thus the theorem is an equivalence of quantum factorization algebras on the boundary. At  $\kappa = 0$  its image contains only  $\alpha$  fields, so its pair contraction vanishes directly.

## 1.7 A COMPACT CALCULATION OF THE CURRENT LEVEL

The quantum differential determines how a product changes when its insertions are moved. A compact two-current calculation fixes the sign before any mode presentation is chosen. For a point  $a$  in a disk, choose a cutoff  $\chi_a$  supported in a small disk about  $a$  and equal to one near its center. The form

$$u_a = -\frac{\bar{\partial}\chi_a}{z-a}$$

is smooth and compactly supported. Stokes' theorem on an annulus about  $a$  gives, for every holomorphic function  $F$  on the disk,

$$\int^{\text{res}} u_a \wedge F(z) dz = F(a).$$

Thus  $u_a$  is a smooth representative of evaluation at  $a$ .

Choose distinct points  $a, b$ , disjoint small disks about them, and cutoffs  $\chi_a, \chi_b$  as above. Choose a common radial cutoff  $\chi(z) = \psi(|z|^2)$ , equal to one on both small disks and supported in a larger disk centered at zero. Put

$$e_a = -\frac{\bar{\partial}\chi}{z-a}, \quad e_b = -\frac{\bar{\partial}\chi}{z-b}, \quad f = \frac{\chi - \chi_a}{z-a}, \quad g = \frac{\chi - \chi_b}{z-b}.$$

The functions  $f, g$  are smooth and compactly supported: the apparent poles are removable. Direct differentiation gives  $\bar{\partial}f = u_a - e_a$  and  $\bar{\partial}g = u_b - e_b$ . On the support of  $u_b$ ,  $f(z) = (z-a)^{-1}$ . Integration by parts and the evaluation formula give

$$\mu_\kappa(f, u_b) = -\kappa \int^{\text{res}} u_b \wedge \partial f = -\kappa f'(b) = \frac{\kappa}{(a-b)^2}.$$

The other replacement has zero contraction. On the common cutoff annulus,  $g = \chi/(z-b)$  and

$$e_a \wedge \partial g = -\frac{\bar{\partial}\chi \wedge \partial\chi}{(z-a)(z-b)} + \frac{\chi \bar{\partial}\chi \wedge dz}{(z-a)(z-b)^2}.$$

Relative to  $d\bar{z} \wedge dz$ , its coefficient is

$$-\frac{(\psi')^2 |z|^2}{(z-a)(z-b)} + \frac{\psi \psi' z}{(z-a)(z-b)^2}.$$

Both Laurent expansions have strictly negative angular Fourier powers, beginning with  $z^{-2}$ . Their angular integrals vanish, so  $\mu_\kappa(e_a, g) = 0$ .

The function generators have degree  $-1$  and the form generators have degree zero. With the specified positive unary differential,  $D = \bar{\partial} + \hbar \Delta_{\mu_\kappa}$ , the Leibniz signs and these contractions give the compact chain relation

$$D(f u_b + e_a g) = u_a u_b - e_a e_b + \frac{\hbar \kappa}{(a-b)^2}.$$

Consequently the actual separated product satisfies

$$[u_a u_b] = [e_a e_b] - \frac{\hbar \kappa}{(a-b)^2} \mathbf{1}.$$

The radial term has the regular expansion  $e_a = \sum_{n \geq 0} a^n u_n$ , where  $u_n = -\bar{\partial}\chi/z^{n+1}$ , and similarly for  $e_b$ . These series converge in every smooth seminorm on the outer annulus. They cannot contribute a diagonal pole. The current determined by the fixed state  $q_1 = [u_0]$  therefore has parameter  $-\kappa$ :

$$J(a)J(b) \sim -\frac{\hbar \kappa}{(a-b)^2}.$$

The remaining problem is to obtain every field and regular product from the supported complex, keeping this normalization.

A direct annular evaluation of the local cocycle has a different role. For holomorphic Laurent functions  $F, G$  and a cutoff equal to one at the inner boundary and zero at the outer boundary,

$$\mu_\kappa(F, \bar{\partial}\chi G) = -\kappa \int^{\text{res}} \bar{\partial}\chi \wedge G dF = \kappa \text{Res } G dF.$$

Only one factor needs compact support in this integral. The minus sign in the separated product comes from moving the quantum contraction to the other side of an exact chain relation. Thus the annular scalar evaluation by itself does not identify the extracted mode action. Multiplying all currents by  $i$  changes their parameter, but also changes the prescribed state and its jet representatives.

### I.8 CAUCHY CONTRACTION ON SMOOTH OBSERVABLES

The quantum differential can be removed by an explicit linear map, provided its effect on products is retained. For  $u \in \Omega_c^{\circ,1}(U)$ , extend  $u$  by zero to  $\mathbb{C}$  and define its Cauchy transform by

$$(\mathcal{C}u)(z) = \frac{1}{2\pi i} \int_{\mathbb{C}} \frac{u(w) \wedge dw}{z - w}.$$

The kernel  $1/(z - w)$  is locally integrable. Stokes' theorem on a small circle gives  $\partial_{\bar{z}}(1/(\pi z)) = \delta_0$ ; the normalization  $d\bar{w} \wedge dw = 2i dA_w$  then gives

$$\bar{\partial}\mathcal{C}u = u, \quad \mathcal{C}\bar{\partial}f = f \quad (f \in C_c^\infty(\mathbb{C})).$$

For the second identity, the first implies that the difference is holomorphic. Integration by parts shows that all moments of  $\bar{\partial}f$  vanish, so expansion of the Cauchy kernel outside a disk containing the support makes  $\mathcal{C}\bar{\partial}f$  compactly supported. The difference is therefore a compactly supported holomorphic function and is zero. The output  $\mathcal{C}u$  is generally not compactly supported; it is used below only inside a scalar integral.

On two degree-zero observable generators, define

$$P_\kappa(u, v) = -\kappa \int^{\text{res}} (\mathcal{C}u) \partial v.$$

Define  $P_\kappa$  to be zero on all other generator-degree pairs. The derivative of  $v$  has compact support, so the displayed integral exists. If  $u = a(w)d\bar{w}$  and  $v = b(z)d\bar{z}$ , it is

$$P_\kappa(u, v) = \frac{\kappa}{\pi^2} \int_{\mathbb{C}^2} \frac{a(w) \partial_z b(z)}{z - w} dA_z dA_w.$$

This formula also defines a pair contraction on an arbitrary compactly supported smooth kernel, by differentiating in the paired  $z$  variable and integrating both paired variables. The output is a smooth kernel on the remaining variables, supported in the compact projection of the input support. For each fixed support  $K$  and multi-index  $\alpha$  in the remaining variables,

$$\|\partial^\alpha \Delta_P F\|_\infty \leq C_{K,\alpha} \|F\|_{C^{|\alpha|+1}}.$$

Indeed, the integral of  $|z - w|^{-1}$  on a fixed compact pair of planar variables is finite. Derivatives in the remaining variables pass under that integral. Thus the contraction is continuous in the smooth-kernel convention already specified; no singular observable or extra completion is introduced.

The pairing  $P_\kappa$  is symmetric. After interchanging  $z, w$ , its antisymmetric part is proportional to

$$\int_{\mathbb{C}^2} \frac{(\partial_z + \partial_w)(a(w)b(z))}{z - w} dA_z dA_w.$$

In difference and translation coordinates, the derivative in the numerator differentiates only the translation coordinate. Its integral is zero by compact support, without differentiating the singular difference kernel.

Let  $K = \Delta_{P_\kappa}$  sum unordered pair contractions, with the shifted Koszul signs, and write  $d = \bar{\partial}$ . The operator  $K$  has degree zero and lowers field number by two. Its commutators are

$$[d, K] = -\Delta_{\mu_\kappa}, \quad [K, \Delta_{\mu_\kappa}] = 0.$$

To verify the first equation, it suffices first to use a function and a form generator:

$$[d, K](fu) = -P_\kappa(\bar{\partial}f, u) = -\mu_\kappa(f, u).$$

The other quadratic generator degrees give zero. Both sides are constant pair-contraction operators, so the identity holds on all monomials. For the second equation, overlapping contractions vanish because the shared factor disappears; disjoint contractions commute because  $K$  is even. The same identities hold on smooth kernels by continuity. One may verify density of decomposable kernels on product boxes by Fourier expansion after inserting compact cutoffs: rapid decay of Fourier coefficients gives convergence in every derivative seminorm. A finite partition of unity on the input support gives the general case.

**Proposition 1.5** (Finite quantum conjugation). *For every open  $U \subseteq \mathbb{C}$  and every  $\kappa \in \mathbb{C}$ , the map*

$$N_U = e^{-\hbar K} : (\mathcal{B}_\kappa(U), D) \longrightarrow (\text{Sym}(\Omega_c^{\circ, \bullet}(U)[1])[\hbar], d)$$

*is a chain isomorphism, with inverse  $e^{\hbar K}$ . It commutes with extension by zero, translations and rotations.*

*Proof.* The two commutators give

$$e^{\hbar K} d e^{-\hbar K} = d + \hbar[K, d] = d + \hbar\Delta_{\mu_\kappa} = D.$$

On field number  $r$ ,  $K^j = 0$  for  $2j > r$ , so all exponentials terminate. They preserve the polynomial complexes and their filtration by field number plus  $\hbar$ -power. The global Cauchy formula uses the same zero extension for every open set. Its kernel and measure are translation invariant and transform compatibly under rotations. These facts prove the stated naturality.  $\square$

The map  $N_U$  is not asserted to preserve multiplication at overlapping supports. Its cross contractions carry the quantum products.

### 1.9 THE VACUUM STATES AND THEIR SMOOTH REPRESENTATIVES

Let  $D_R = \{|z| < R\}$  and choose a radial cutoff  $\chi$  equal to one near zero and zero near the boundary. For  $n \geq 0$  put

$$u_n = -\frac{\bar{\partial}\chi}{z^{n+1}}.$$

The support avoids zero. Stokes' theorem gives its divided-jet moments:

$$\int_{D_R}^{\text{res}} u_n \wedge F(z) dz = \frac{F^{(n)}(0)}{n!}, \quad M_j(u_n) = \delta_{jn}, \quad M_j(u) = \int_{D_R}^{\text{res}} u \wedge z^j dz.$$

A compactly supported holomorphic function vanishes. A compact  $(0, 1)$ -form with every holomorphic moment zero has a compact primitive: its Cauchy transform vanishes outside a disk containing its support, by expansion in  $w/z$ . Conversely, integration by parts kills the moments of  $\bar{\partial}f$  for compact  $f$ .

The required statement on completed tensors follows from an explicit estimate. If the input is supported in  $|z| \leq r_o < R$ , choose the cutoff annulus with inner radius  $\sigma > r_o$ . Then

$$|M_n(u)| \leq C r_o^n, \quad \|u_n\|_{C^d} \leq C_d (n+1)^{d+1} \sigma^{-n-1}.$$

Thus the maps

$$\Pi u = \sum_{n \geq 0} M_n(u) u_n, \quad h_D u = \mathcal{C}(u - \Pi u)$$

are well defined on this fixed support space, with convergence in all derivative seminorms and compact output. On functions set  $\Pi = h_D = 0$ . The moments and the Cauchy identities prove

$$dh_D + h_D d = 1 - \Pi.$$

For a compact smooth kernel on  $D_R^r$ , choose a common inner support radius and a larger cutoff annulus in each variable. The moment bounds multiply, so the resulting multiple series converge in every mixed derivative seminorm. With shifted tensor signs, the map

$$H_r = \sum_{j=1}^r \Pi^{\otimes(j-1)} \otimes h_D \otimes 1^{\otimes(r-j)}$$

satisfies  $dH_r + H_r d = 1 - \Pi^{\otimes r}$  by telescoping. Each variable is treated once. No single cutoff operator on all compact supports is required. Averaging over the finite symmetric group proves the same assertion for the completed symmetric powers.

The tensor cohomology is consequently in top antiholomorphic degree  $r$ , hence in shifted cohomological degree zero, and is determined by its multivariable moments. This computes the odd generators by their differential; it does not discard them.

Rotations act on the representatives by

$$R_\theta^* u_n = e^{-i(n+1)\theta} u_n.$$

Define the state weight as minus this pullback rotation weight. The radial homotopies commute with rotations, including simultaneous rotation of all variables. At state weight  $N$ , a surviving tensor must satisfy

$$\sum_{i=1}^r (n_i + 1) = N.$$

There are finitely many such tuples, and  $r \leq N$ . Thus the argument applies to a fixed diagonal weight without assuming a finite Fourier expansion in the individual variables.

Rotation invariance also gives  $P_\kappa(u_n, u_m) = 0$  for all  $n, m$ : the two inputs have strictly positive total state weight, while a scalar has weight zero. The map  $N_{D_R}$  therefore fixes every monomial in these radial representatives. Combining its chain isomorphism with the moment calculation gives

$$V_\kappa = \mathbb{C}[q_1, q_2, \dots][\hbar], \quad q_{n+1} \mapsto [u_n], \quad |q_m| = 0, \quad \text{wt}(q_m) = m.$$

More precisely,  $V_\kappa$  is the direct sum of finite-weight cohomology spaces of  $\mathcal{B}_\kappa(D_R)$ . Each weight space is finite free over  $\mathbb{C}[\hbar]$ , and there is no odd cohomology or  $\hbar$ -torsion. The full disk cohomology also contains infinite moment sequences and is not identified with this direct sum.

Polynomial multiplication specifies the normal state basis  $\prod q_{m_i} \mapsto [\prod u_{m_i-1}]$ . It does not assert that multiplication at overlapping supports descends to quantum cohomology. Changing the radial cutoff or enlarging the centered disk preserves this basis: applying  $N$  reduces the comparison to the equality of its multivariable moments.

#### 1.10 TRANSPORT OF EVERY SEPARATED PRODUCT

For disjoint opens  $U_1, \dots, U_s \subseteq U$ , the original factorization product is extension by zero followed by multiplication  $m$ . Its mixed  $\mu_\kappa$  contractions vanish because the supports are disjoint. Under Cauchy conjugation its exact formula is

$$N_U m(N_{U_1}^{-1} \otimes \dots \otimes N_{U_s}^{-1}) = m \exp \left( -\hbar \sum_{i < j} \Delta_{P_\kappa}^{ij} \right),$$

where  $\Delta_{P_\kappa}^{ij}$  contracts a pair with one factor in input  $i$  and one in input  $j$ . To prove the formula, decompose  $K$  on a product into contractions internal to each input and contractions between inputs. All these even operators commute. The internal exponentials cancel  $N_{U_i}^{-1}$ , leaving exactly the displayed cross contractions.

The exponential terminates on every input. Its terms are matchings of the individual field factors, with edges joining different input groups. A matching with  $j$  edges appears  $j!$  times, one for each ordering of its disjoint contractions; the exponential coefficient  $1/j!$  cancels that multiplicity. The proof extends to all compact smooth kernels by the continuity estimate above. It commutes with nesting, either by conjugating the original factorization maps or by observing that each nested expression contains each cross pair exactly once.

For a small radial cutoff at  $a$ , write

$$u_{a,n} = -\frac{\bar{\partial} \chi_a}{(z-a)^{n+1}}, \quad n \geq 0.$$

Its moments are divided derivatives at  $a$ . Outside the supporting disk,

$$(\mathcal{C} u_{a,n})(z) = \frac{1}{(z-a)^{n+1}}.$$

This follows by applying the moment formula to the holomorphic function  $w \mapsto (z-w)^{-1}$ . For disjoint supporting disks at  $a, b$ , integration by parts therefore gives

$$\begin{aligned} P_\kappa(u_{a,n}, u_{b,m}) &= -\frac{\kappa}{m!} \partial_z^{m+1} (z-a)^{-n-1} \Big|_{z=b} \\ &= \frac{\kappa}{n!m!} \partial_a^n \partial_b^m (a-b)^{-2}. \end{aligned}$$

Put  $\lambda = -\hbar \kappa$ . The actual cross contraction of divided jets of orders  $n, m$  is

$$C_{n,m}(a, b) = \lambda \frac{(-1)^n (n+m+1)!}{n!m!} (a-b)^{-n-m-2}.$$

In particular,  $C_{0,0}(a, b) = \lambda(a-b)^{-2}$ , in agreement with the compact two-current calculation.

Holomorphic motion of the remaining state is equally explicit. A common outer radial cutoff gives

$$-\frac{\bar{\partial} \chi}{(z-a)^{n+1}} = \sum_{k \geq 0} \binom{n+k}{k} a^k u_{n+k}.$$

The difference from  $u_{a,n}$  has zero moments and a compact primitive. For products of co-centered representatives, all internal  $P$  terms vanish by rotation about that center. The same comparison therefore holds for each polynomial state. It gives the translation derivation

$$Tq_m = mq_{m+1}, \quad T1 = 0.$$

The moving state corresponding to  $q_m$  is  $e^{aT}q_m$ .

For arbitrary polynomial states inserted at distinct points, label each of their jet factors by  $\alpha$ , with jet order  $r_\alpha$  and insertion group  $i(\alpha)$ . The actual product, expressed by  $N$  and the moment coordinates, is

$$\sum_M \left( \prod_{\{\alpha, \beta\} \in M} C_{r_\alpha, r_\beta}(a_{i(\alpha)}, a_{i(\beta)}) \right) \left( \prod_{\gamma \text{ unpaired}} e^{a_{i(\gamma)}T} q_{r_\gamma+1} \right).$$

Here  $M$  ranges over matchings joining different insertion groups. There are finitely many matchings, including the empty matching. The empty matching retains the regular terms. For fixed numerical insertion points, the unpaired series have smooth representatives on a common outer annulus and converge in all derivative seminorms, by the geometric moment bounds. Thus the formula is an equality in full disk cohomology. Taking weight coefficients and radial Laurent coefficients of this equality gives polynomial states. This is the extraction from the smooth carrier.

## I.II COMPACT HOMOTOPIES FOR NORMAL ORDERING

The Wick expansion gives a cohomology class of supported observables. A chain comparison also needs a compact primitive for its difference from the original product. At three insertions, that primitive itself has a quantum correction. A finite Koszul complex inside the boundary observables constructs these primitives in every polynomial degree.

Fix a disk  $U = D_R \subset \mathbb{C}$  and finitely many distinct points in  $U$ . Choose disjoint centered disks about these points, with radial cutoffs  $\chi_a$  equal to one near their centers. Choose a radial cutoff  $\chi$  about zero which equals one on a centered closed disk containing all the smaller closed disks. All cutoffs have compact support in their stated disks. Choose a finite list of distinct pairs  $(a_i, n_i)$ , where  $n_i \geq 0$  is a divided holomorphic jet order, and index it by  $1 \leq i \leq N$ . Several pairs may have the same point. Set

$$u_i = -\frac{\bar{\partial}\chi_{a_i}}{(z - a_i)^{n_i+1}}, \quad e_i = -\frac{\bar{\partial}\chi}{(z - a_i)^{n_i+1}}, \quad f_i = \frac{\chi - \chi_{a_i}}{(z - a_i)^{n_i+1}}.$$

The apparent poles are removable. All three expressions are smooth and compactly supported in  $U$ , and  $\bar{\partial}f_i = u_i - e_i$ . In the boundary observable complex,  $u_i, e_i$  have degree zero, while  $f_i$  has degree  $-1$ .

Keep the bulk parameter  $\kappa$  and the positive differential  $D = \bar{\partial} + \hbar\Delta_{\mu_\kappa}$  fixed. Define the symmetric constants

$$p_{ij} = \begin{cases} \frac{\kappa}{n_i!n_j!} \partial_{a_i}^{n_i} \partial_{a_j}^{n_j} (a_i - a_j)^{-2}, & a_i \neq a_j, \\ 0, & a_i = a_j. \end{cases}$$

In particular,  $p_{ii} = 0$ . The extracted current has level  $-\kappa$ , so its Wick contraction is  $-\hbar p_{ij}$ .

**Lemma 1.6** (Contractions of compact representatives). *The Cauchy pairing and the quantum cocycle satisfy*

$$\begin{aligned} P_\kappa(u_i, u_j) &= p_{ij}, & P_\kappa(e_i, u_j) &= P_\kappa(e_i, e_j) = 0, \\ \mu_\kappa(f_i, u_j) &= p_{ij}, & \mu_\kappa(f_i, e_j) &= 0. \end{aligned}$$

*The pairing  $P_\kappa$  is zero on pairs involving a function generator. The cocycle  $\mu_\kappa$  is zero on two function generators or two form generators.*

*Proof.* For distinct centers, the separated Cauchy calculation gives the displayed derivative formula for  $P_\kappa(u_i, u_j)$ . For a common center, simultaneous rotation multiplies this pairing by weight  $-(n_i + n_j + 2)$ . The scalar output has weight zero. Rotation invariance therefore makes the pairing zero.

The Cauchy transform of an outer representative is

$$\mathcal{C}e_i = \frac{1 - \chi}{(z - a_i)^{n_i+1}}.$$

Both sides have Dolbeault derivative  $e_i$  and tend to zero at infinity. Their difference is an entire function tending to zero, so it vanishes. The displayed expression is zero near the support of every  $u_j$ . Hence  $P_\kappa(e_i, u_j) = 0$ . On the common annulus, each  $e_i$  has the convergent expansion

$$e_i = \sum_{r \geq 0} \binom{n_i + r}{r} a_i^r \left( -\frac{\bar{\partial}\chi}{z^{n_i+r+1}} \right).$$

Each term has strictly positive state weight. Pairing two such terms gives zero by rotation invariance. Smooth convergence and continuity of  $P_\kappa$  give  $P_\kappa(e_i, e_j) = 0$ .

The Cauchy identity on compact functions gives

$$\mu_\kappa(f_i, v) = P_\kappa(\bar{\partial}f_i, v) = P_\kappa(u_i - e_i, v)$$

for each form generator  $v$ . This proves both cocycle formulas. The remaining vanishing statements follow from their specified degrees.  $\square$

Let  $x_i, y_i, \theta_i$  denote the observable generators represented by  $u_i, e_i, f_i$ , respectively. The symbols  $f_i$  continue to denote the actual compact functions. Thus

$$|x_i| = |y_i| = 0, \quad |\theta_i| = -1, \quad Q = \mathbb{C}[x_1, \dots, x_N, y_1, \dots, y_N] \otimes \Lambda(\theta_1, \dots, \theta_N)[\hbar].$$

The odd variables anticommute. Use left derivatives in them, so  $\partial_{\theta_i}(\theta_{j_1} \cdots \theta_{j_s})$  has sign  $(-1)^{r-1}$  when  $i = j_r$  in increasing order. All tensor substitutions use these cohomological degrees. The map  $\rho : Q \rightarrow \mathcal{B}_\kappa(U)$  substitutes the compact representatives and multiplies them as observable polynomials.

**Proposition 1.7** (A quantum complex of compact currents). *The map  $\rho$  embeds  $Q$  as a subcomplex of the full smooth boundary observable complex. Its differential is*

$$D_Q = \sum_i (x_i - y_i) \partial_{\theta_i} + \hbar \sum_{i < j} p_{ij} (\partial_{x_j} \partial_{\theta_i} + \partial_{x_i} \partial_{\theta_j}).$$

*This assertion concerns a complex of vector spaces over  $\mathbb{C}[\hbar]$ . The ordinary multiplication on  $Q$  is not a differential graded multiplication when some  $p_{ij} \neq 0$ .*

*Proof.* The preceding lemma computes every possible contraction of its generators. Together with  $\bar{\partial}f_i = u_i - e_i$ , it gives (1.7) on every polynomial.

The  $u_i$  are linearly independent: their supports separate the centers, and the angular Fourier powers separate the orders at one center. Their supports are disjoint from the common outer annulus. The  $e_i$  are linearly independent there. Indeed, a relation among them would make a rational function  $\sum_i c_i (z - a_i)^{-n_i-1}$  vanish on an open subset of the annulus. It would vanish identically, and its partial fractions give  $c_i = 0$ . Thus all the form generators are independent. A relation among the  $f_i$  would give a relation among  $u_i - e_i$  after differentiation, so all function generators are independent as well. Symmetric and exterior powers of these finite-dimensional subspaces embed

into their smooth tensor powers. For example, continuous dual linear functionals selecting their basis vectors supply a left inverse on every tensor power, and then on its finite-group coinvariants. This proves injectivity of  $\rho$  in the stated smooth-kernel convention.

Finally,  $D_Q(\theta_i x_j) - (D_Q \theta_i) x_j = \hbar p_{ij}$ . This proves the asserted failure of the Leibniz identity whenever the displayed scalar is nonzero.  $\square$

Write  $\delta_i = x_i - y_i$ , let  $A = \mathbb{C}[y_1, \dots, y_N][\hbar]$  with zero differential, and let  $\iota : A \rightarrow Q$  be the inclusion. Let  $\pi : Q \rightarrow A$  set all  $\delta_i, \theta_i$  to zero. On  $Q$  define

$$d_o = \sum_i \delta_i \partial_{\theta_i}, \quad K_Q = \sum_{i < j} p_{ij} \partial_{x_i} \partial_{x_j}, \quad N_Q = e^{-\hbar K_Q}.$$

The derivatives  $\partial_{x_i}$  fix  $y_j$  and  $\theta_j$ . For a monomial with total degree  $q$  in the variables  $\delta_i, \theta_i$ , define

$$h_o = \begin{cases} \frac{1}{q} \sum_i \theta_i \partial_{\delta_i}, & q > 0, \\ 0, & q = 0. \end{cases}$$

Extend this definition linearly over  $A$ . All exponentials below terminate on each polynomial.

**Theorem 1.8** (Compact quantum contraction). *The maps*

$$\Pi = \pi N_Q : Q \longrightarrow A, \quad H = N_Q^{-1} h_o N_Q : Q \longrightarrow Q[-1]$$

*and the inclusion  $\iota$  satisfy*

$$\Pi \iota = 1, \quad D_Q H + H D_Q = 1 - \iota \Pi, \quad H \iota = \Pi H = H^2 = 0.$$

*Both  $\iota$  and  $\Pi$  are chain maps. Consequently  $Q$  has cohomology  $A$  in degree zero and zero cohomology in every other degree. These identities hold for every  $\kappa$ , including zero.*

*Proof.* The odd derivation  $s_o = \sum_i \theta_i \partial_{\delta_i}$  satisfies

$$d_o s_o + s_o d_o = \sum_i \delta_i \partial_{\delta_i} + \sum_i \theta_i \partial_{\theta_i}.$$

One checks the equality on the generators. Both sides are even derivations, which proves it on every polynomial. The right side acts by  $q$  on the specified degree- $q$  summand. Thus  $d_o h_o + h_o d_o = 1 - \iota \pi$ . The operator  $s_o$  preserves  $q$  and has square zero, because the odd factors anticommute while the even derivatives commute. This also gives  $h_o^2 = h_o \iota = \pi h_o = 0$ .

Direct commutation gives

$$[K_Q, d_o] = \sum_{i < j} p_{ij} (\partial_{x_j} \partial_{\theta_i} + \partial_{x_i} \partial_{\theta_j}), \quad [K_Q, [K_Q, d_o]] = 0.$$

Therefore  $D_Q = N_Q^{-1} d_o N_Q$ . Also  $N_Q \iota = \iota = N_Q^{-1} \iota$ , since  $K_Q$  kills polynomials in the  $y_i$  alone. Conjugating the classical contraction gives every equation in (1.8). It also proves  $D_Q^2 = 0$  and the chain-map assertions. No inverse of  $\kappa$  or  $\hbar$  occurs.  $\square$

The inclusion into the full boundary complex preserves these explicit homotopies. For a polynomial  $F(x)$  in the local current generators,

$$F(u) - \iota_U \left( e^{-\hbar K_Q} F(x) \Big|_{x=y} \right) = D \rho H F(x), \quad \iota_U = \rho \iota.$$

Here  $F(u) = \rho F(x)$  and  $D_Q F(x) = 0$ . The equality is in compactly supported observables, before taking cohomology. Distinct input groups have disjoint small disks. Their actual factorization product is the ordinary polynomial on the left. Expanding the exponential on the right gives every matching between groups, with contraction  $-\hbar p_{ij}$ . Pairings within a group vanish by the common-center convention. Repeated factors are counted by the polynomial derivatives.

For example, with  $\delta_i = x_i - y_i$  and three different indices,

$$\begin{aligned} H(x_1 x_2) &= \frac{1}{2} \theta_1(x_2 + y_2) + \frac{1}{2} \theta_2(x_1 + y_1), \\ \Pi(x_1 x_2) &= y_1 y_2 - \hbar p_{12}, \\ H(x_1 x_2 x_3) &= \sum_{i=1}^3 \theta_i \left( y_j y_k + \frac{\delta_j y_k + y_j \delta_k}{2} + \frac{\delta_j \delta_k}{3} - \frac{2\hbar p_{jk}}{3} \right), \\ \Pi(x_1 x_2 x_3) &= y_1 y_2 y_3 - \hbar(p_{12} y_3 + p_{13} y_2 + p_{23} y_1). \end{aligned}$$

In the cubic sum,  $\{j, k\} = \{1, 2, 3\} \setminus \{i\}$ . Its summand is symmetric in  $j, k$ . The classical Euler homotopy contributes the first three terms. Applying  $N_Q$  before it contributes  $-\hbar p_{jk} \theta_i$ . Applying  $N_Q^{-1}$  afterwards contributes  $\hbar p_{jk} \theta_i / 3$ . Their sum gives the printed correction. In particular, omitting that correction leaves the defect

$$\frac{2\hbar}{3} (p_{12} \delta_3 + p_{13} \delta_2 + p_{23} \delta_1)$$

in the cubic chain identity. For nonzero covariance, independence of the variables  $\delta_i$  makes this defect nonzero. The binary identity alone cannot fix it.

**Proposition 1.9** (Split inclusion in smooth boundary observables). *There is an explicit chain map  $L_U : \mathcal{B}_\kappa(U) \rightarrow A$  with  $L_U \iota_U = 1$ . Thus the annular polynomial states inject into the cohomology of the full smooth boundary observable complex. On the subcomplex  $\rho Q$ ,  $L_U \rho = \Pi$ .*

*Proof.* Choose holomorphic polynomials  $F_i(z)$  such that

$$\frac{F_i^{(n_j)}(a_j)}{n_j!} = \delta_{ij} \quad (1 \leq i, j \leq N).$$

Such polynomials exist by finite interpolation. To verify existence, let  $m_a = 1 + \max\{n_j : a_j = a\}$  and  $M = \sum_a m_a$ . The map from polynomials of degree less than  $M$  to their full jets of orders less than  $m_a$  is injective: a polynomial in its kernel is divisible by  $\prod_a (z - a)^{m_a}$ , of degree  $M$ . The domain and target have dimension  $M$ , so the map is surjective. Prescribe the stated entries and set all other jet entries to zero.

On the classical observable complex, send a form generator  $v$  to

$$R(v) = \sum_i \left( \int_U^{\text{res}} v \wedge F_i(z) dz \right) y_i$$

and a function generator to zero. Extend by multiplication. For a compact smooth kernel, integrate in each variable and take the finite sum over its indices  $i$ . These integrals are continuous on each fixed compact support and respect the symmetric coinvariants. Integration by parts and holomorphicity give  $Rd = 0$ . The moments of both  $u_j$  and  $e_j$  give  $R(u_j) = R(e_j) = y_j$ . Thus  $R\rho = \pi$  on the classical complexes.

The Cauchy contraction restricts to  $K_Q$  on  $\rho Q$ , by Lemma 1.6. Consequently  $N_U \rho = \rho N_Q$ . Set  $L_U = R N_U$ , using the chain isomorphism of Proposition 1.5. Then  $L_U \rho = \pi N_Q = \Pi$  and  $L_U \iota_U = 1$ .  $\square$

For a fixed set of centers, increasing the finite jet list preserves  $D_Q, K_Q, h_o, \Pi, H$  on the smaller polynomial algebra. Hence the compact contraction also holds on the union of all finite jet lists. Its elements still have finite polynomial degree. The interpolation projection  $L_U$  requires a finite list and is not asserted to be compatible with this union. The union has more jet generators, but it does not include arbitrary smooth kernels or all infinite moment sequences.

The maps  $D_Q, N_Q, H, \Pi$  are polynomial in  $\hbar$  on each input polynomial. They therefore extend to  $\widehat{Q} = \varprojlim_m Q/\hbar^m Q$  and  $\widehat{A} = \varprojlim_m A/\hbar^m A$ , where each coefficient has finite polynomial degree. A fixed coefficient of an exponential receives finitely many terms. The same identities hold modulo every  $\hbar^m$ , hence in the inverse limit. This is a contraction of the specified completed subcomplex. It does not compute the cohomology of the completed full boundary complex.

The compact collar chain map sends  $\rho HF$  to a primitive for the corresponding bulk–boundary product difference. Its chain identity follows by applying that map to (1.11); its separated-product identity identifies the first term with the actual collar product. Thus the normalization  $-\hbar\kappa(a-b)^{-2}$  and its compact primitives pass through the same boundary comparison. Construction of operations on collision chains requires further maps compatible with those chains and their boundary strata.

## 1.12 FIELDS, MODES AND REGULAR PRODUCTS

Applying the separated-product formula to one current and a monomial at zero gives the operator-valued formal distribution

$$J(z) = \sum_{m \geq 1} q_m z^{m-1} + \lambda \sum_{m \geq 1} m \partial_{q_m} z^{-m-1}.$$

The first sum is the creation part  $J_-(z)$ , acting by multiplication; the second is the annihilation part  $J_+(z)$ , acting by derivations. A mode  $a_{(n)}$  is the coefficient of  $z^{-n-1}$  in its field. For this current,

$$J_{-m} = q_m, \quad J_m = -\hbar\kappa m \partial_{q_m} \quad (m > 0), \quad J_0 = 0.$$

These operators were obtained from compact insertions and their quantum differential with the fixed identification  $q_{n+1} = [u_n]$ .

Write  $J^{[r]}(z) = \partial_z^r J(z)/r!$  for its divided derivatives. For a monomial state  $a = \prod_{i=1}^r q_{m_i}$ , its field is

$$Y(a, z) =: \prod_{i=1}^r J^{[m_i-1]}(z) :.$$

Normal ordering here has an explicit definition. Split each divided current into creation and annihilation parts, expand the finite product, and put every multiplication operator to the left of every derivation. Both types commute within themselves, so the answer is independent of the ordering of the factors of  $a$ . Expanding those derivations against any polynomial state gives precisely the preceding matching formula with the second insertion at zero. Thus  $Y$  is the extracted state-field operation, including its regular coefficients.

For example, the complete two-jet product is

$$\begin{aligned} Y(q_m, z)q_n &= \sum_{k \geq 0} \binom{m+k-1}{k} q_{m+k} q_n z^k \\ &\quad + \lambda \frac{(-1)^{m-1} (m+n-1)!}{(m-1)!(n-1)!} z^{-m-n}. \end{aligned}$$

For a quadratic state, the regular coefficient contains a contraction with a translated unpaired current:

$$(q_1^2)_{(-1)}q_1 = q_1^3 + 2\lambda q_3 = q_1^3 - 2\hbar\kappa q_3.$$

Indeed,  $Y(q_1^2, z)q_1 = J_-(z)^2 q_1 + 2\lambda z^{-2} J_-(z)$ . Since  $q_{1(-1)}q_1 = q_1^2$  and  $q_{1(-1)}q_1^2 = q_1^3$ , this also gives the regular-product associator

$$(q_{1(-1)}q_1)_{(-1)}q_1 - q_{1(-1)}(q_{1(-1)}q_1) = 2\lambda q_3.$$

Polynomial multiplication in the normal state basis and the regular vertex product are therefore different operations at nonzero level.

Every field is lower truncated on every polynomial state: only finitely many annihilation modes can differentiate that state. At any fixed remaining nonnegative power of  $z$ , its creation factors have only finitely many index choices. All field coefficients are therefore well-defined polynomial endomorphisms.

### 1.13 ALL OPERATOR PRODUCTS AND THE VERTEX IDENTITIES

Direct differentiation on the polynomial states gives

$$[J_m, J_n] = \lambda m \delta_{m+n,0}, \quad [J_+(z), J_-(w)] = \iota_{z,w} \frac{\lambda}{(z-w)^2},$$

where  $\iota_{z,w}$  means expansion in  $w/z$ . Commuting annihilation factors past creation factors in the normal fields proves their complete Wick formula. If the factors of  $a$  have jet orders  $r_\alpha$  and those of  $b$  have jet orders  $s_\beta$ , then

$$Y(a, z)Y(b, w) = \sum_M \iota_{z,w} \left( \prod_{\{\alpha, \beta\} \in M} C_{r_\alpha, s_\beta}(z, w) \right) : \prod_{\alpha \text{ unpaired}} J^{[r_\alpha]}(z) \prod_{\beta \text{ unpaired}} J^{[s_\beta]}(w) : .$$

Each matching joins a factor of  $a$  to a factor of  $b$ ; unpaired fields keep their original positions and divided derivatives. Every commutation removes a paired creation and annihilation factor, so induction on the number of factors proves the formula and its matching multiplicities. It also proves the analogous formula for any number of insertion groups. For instance,

$$Y(q_1^2, z)Y(q_1^2, w) = J(z)^2 J(w)^2 : + \frac{4\lambda}{(z-w)^2} : J(z)J(w) : + \frac{2\lambda^2}{(z-w)^4},$$

with each rational coefficient expanded by  $\iota_{z,w}$ . Taylor expansion of the unpaired factors gives every regular as well as every singular operator-product coefficient.

Set  $\mathbf{1} = \mathbf{1}$ . The annihilation operators kill the vacuum, and the creation part of  $J^{[m-1]}(z)$  is  $e^{zT} q_m$ . Hence

$$Y(\mathbf{1}, z) = \text{id}, \quad Y(a, z)\mathbf{1} = e^{zT} a, \quad T\mathbf{1} = 0.$$

The translation identities follow from

$$[T, q_m] = m q_{m+1}, \quad [T, \partial_{q_m}] = -(m-1) \partial_{q_{m-1}},$$

where the second expression is zero at  $m = 1$ . They give

$$[T, Y(a, z)] = \partial_z Y(a, z) = Y(Ta, z).$$

Locality also follows from the same matching formula. The reverse ordered product has the same normally ordered unpaired factors and the reverse expansion of the same rational contractions. If  $a, b$  have  $r, s$  factors, respectively, put

$$L = \sum_{\alpha} r_{\alpha} + \sum_{\beta} s_{\beta} + 2 \min(r, s).$$

Every matching denominator divides  $(z - w)^L$ . After multiplication by this factor the two radial expansions agree, proving  $(z - w)^L [Y(a, z), Y(b, w)] = 0$ .

The iterate identity requires compatibility of the regular terms as well. Introduce a formal displacement  $x$  and put  $z = w + x$ . Each unpaired jet expands as

$$J^{[r]}(w + x) = \sum_{k \geq 0} \binom{r+k}{k} x^k J^{[r+k]}(w).$$

These are exactly the translated jet states in  $Y(a, x)b$ . For any third polynomial state  $c$ , matchings between the three groups  $a, b, c$ , at  $z, w, 0$ , define one expression

$$F_{a,b,c}(z, w) \in V_{\kappa}[[z, w]][z^{-1}, w^{-1}, (z - w)^{-1}]$$

with finite pole orders. Its three expansions are

$$\iota_{z,w} F = Y(a, z)Y(b, w)c,$$

$$\iota_{w,z} F = Y(b, w)Y(a, z)c,$$

$$\iota_{w,x} F(w + x, w) = Y(Y(a, x)b, w)c.$$

For the last identity, group a matching first by its  $a$ - $b$  edges and then by its edges to  $c$ . This retains each matching and its multiplicity. At fixed output weight, the expression is a rational function with polynomial numerator in a finite free weight space. This makes all three expansions legitimate coefficient calculations.

For completeness, Jacobi follows from an elementary formal identity. Write  $\delta(t) = \sum_{n \in \mathbb{Z}} t^n$  and  $\delta(z, w) = z^{-1} \delta(w/z)$ . With the corresponding binomial expansion orders,

$$x^{-1} \delta\left(\frac{z-w}{x}\right) - x^{-1} \delta\left(\frac{w-z}{-x}\right) = w^{-1} \delta\left(\frac{z-x}{w}\right).$$

The coefficients of negative powers of  $x$  cancel on the left, because they come from nonnegative powers of  $z - w$ . For  $k \geq 0$ , the coefficient of  $x^k$  on each side is  $\partial_w^k \delta(z, w)/k!$ ; this is the binomial identity  $\binom{-l-1}{k} = (-1)^k \binom{l+k}{k}$ . Multiply the identity by each rational monomial of a fixed-weight component of  $F_{a,b,c}$ , using  $z = w + x$  in its collision expansion. The same binomial calculation, including negative exponents, gives

$$\begin{aligned} x^{-1} \delta\left(\frac{z-w}{x}\right) Y(a, z)Y(b, w) - x^{-1} \delta\left(\frac{w-z}{-x}\right) Y(b, w)Y(a, z) \\ = w^{-1} \delta\left(\frac{z-x}{w}\right) Y(Y(a, x)b, w). \end{aligned}$$

This is the full vertex Jacobi identity. In modes it reads, for all integers  $m, n, k$ ,

$$\sum_{j \geq 0} \binom{m}{j} (a_{(n+j)} b_{(m+k-j)}) = \sum_{j \geq 0} (-1)^j \binom{n}{j} (a_{(m+n-j)} b_{(k+j)}) - (-1)^n b_{(n+k-j)} a_{(m+j)}.$$

The sums are locally finite on polynomial states by lower truncation. All extracted states are even, as the cohomology calculation proved.

**Theorem 1.10** (The boundary current at every level). *For every  $\kappa \in \mathbb{C}$ , the polynomial finite-weight chiral vacuum extracted from  $\mathcal{B}_\kappa$  is a vertex algebra over  $\mathbb{C}[\hbar]$ . Its state space, translation and fields are*

$$V_\kappa = \mathbb{C}[q_1, q_2, \dots][\hbar], \quad Tq_m = mq_{m+1}, \quad Y\left(\prod_i q_{m_i}, z\right) =: \prod_i J^{[m_i-1]}(z) ;,$$

where  $J_{-m} = q_m$ ,  $J_m = -\hbar\kappa m \partial_{q_m}$  for  $m > 0$ , and  $J_0 = 0$ . The identification  $q_{n+1} = [u_n]$  preserves every mode, divided jet, regular product and separated factorization product. Its Heisenberg level is  $-\kappa$  with this fixed current.

*Proof.* The continuous Cauchy conjugation and compact moment homotopies give the stated cohomology and its normal state basis. Transport through that conjugation gives the actual separated-product matching formula. Expanding it with one input at zero yields exactly the fields above, including their regular terms. The preceding vacuum, translation, locality and Jacobi calculations prove all vertex identities. The chain relation and the mode calculation both give level  $-\kappa$  without changing the chosen state representatives.  $\square$

**Theorem 1.11** (The zero-level current). *At  $\kappa = 0$ , the extracted vertex algebra is the commutative differential algebra  $\mathbb{C}[q_1, q_2, \dots][\hbar]$  with  $Tq_m = mq_{m+1}$  and  $Y(a, z)b = (e^{zT}a)b$ . It agrees with the algebraic zero-level Heisenberg vacuum in every mode and product.*

*Proof.* Both quantum contractions and the annihilation part of  $J$  vanish at zero. Only the empty matching remains. Its state-field formula is precisely  $(e^{zT}a)b$ .  $\square$

**Corollary 1.12** (Augmentation of the extracted current). *The zero-level polynomial vertex algebra has the augmentation  $q_m \mapsto 0$  for all  $m \geq 1$ . At nonzero level there is no augmentation to  $\mathbb{C}[\hbar]$  as a vertex algebra over  $\mathbb{C}[\hbar]$ .*

*Proof.* At zero, evaluation at  $q_m = 0$  preserves multiplication, translation and the vacuum. At nonzero level  $q_{1(1)}q_1 = -\hbar\kappa 1$ . A vertex map to the scalar vertex algebra would send its left side to zero and its right side to a nonzero scalar.  $\square$

#### 1.14 FORMAL COMPLETION AND THE BULK-BOUNDARY COMPARISON

The preceding constructions are polynomial in  $\hbar$  and in field number. The Cauchy exponentials terminate on every observable, and every fixed tuple of polynomial states has finitely many matchings and finite pole orders. Smooth moment expansions converge on a common outer annulus before their finite-weight coefficients are extracted.

Only after this polynomial extraction take the coefficientwise  $\hbar$ -completion

$$\widehat{V}_\kappa = \varprojlim_M V_\kappa / \hbar^M V_\kappa = \mathbb{C}[q_1, q_2, \dots][[\hbar]].$$

Every coefficient of  $\hbar$  is a finite polynomial, while its field number and weight may increase with that coefficient. The completed state-field operation has target

$$\varprojlim_M (V_\kappa / \hbar^M V_\kappa)((z)).$$

Modulo each  $\hbar^M$ , the inputs are polynomial states and all the preceding identities hold. The operations commute with reduction in  $M$ . Vacuum, translation, Jacobi and locality modulo each  $\hbar^M$  therefore pass to this inverse limit.

There need not be a uniform Laurent pole bound for completed inputs. At  $\kappa \neq 0$ , take  $a = \sum_{n \geq 0} \hbar^n q_{n+1}$  and  $b = q_1$ . Its singular terms are

$$-\kappa \sum_{n \geq 0} (-1)^n (n+1) \hbar^{n+1} z^{-n-2}.$$

The pole orders increase with the  $\hbar$ -order. Thus the inverse limit above is the correct field target, rather than  $\widehat{\mathcal{V}}_\kappa((z))$  with a single Laurent bound. This passage completes the extracted state and field algebra; it makes no claim that cohomology commutes with completion of the entire factorization algebra.

The supported quantum collar map of Theorem 1.4 has its original fields, differential and contraction. It commutes with translations, rotations and every separated product. Hence the current states in the bulk–boundary pushforward are

$$q_m \mapsto [I_\kappa(u_{m-1})], \quad \prod_i q_{m_i} \mapsto \left[ \prod_i I_\kappa(u_{m_i-1}) \right].$$

Preservation of separated products passes to their moving-point coefficients, so this comparison preserves every extracted field and mode. The nonzero BF–Chern–Simons field change has bulk parameter  $\kappa$ , whereas the fixed boundary current has level  $-\kappa$  with the printed positive unary differential. At zero, this same supported construction realizes the commutative current while retaining the additional bulk gauge and field cohomology.

### 1.15 SEPARATED PRODUCTS AND TOPOLOGICAL HIGHER OPERATIONS

For distinct holomorphic insertion points, the zero-level Wick product has no contraction terms. Its normal-ordering map sends

$$x_{i,r} \mapsto \partial_{z_i}^r J(z_i), \quad x_{1,0} \star x_{2,0} \star x_{3,0} = x_{1,0} x_{2,0} x_{3,0}.$$

These are equalities on the actual boundary field representation from Theorem 1.11.

An auxiliary topological interval gives the same higher operations on these fields as on their algebraic presentation. To specify the statement, let  $e_0, e_1$  be endpoint cochains and  $\eta$  the oriented edge, with  $de_0 = -\eta$ ,  $de_1 = \eta$ . Their polynomial-form inclusion and homotopy are

$$i(e_0) = 1 - t, \quad i(e_1) = t, \quad i(\eta) = dt, \quad h(gdt) = \int_0^t g - t \int_0^1 g.$$

Projection takes endpoint values and the integral of a one-form. Direct differentiation gives  $dh + hd = 1 - i p$  and  $hi = ph = h^2 = 0$ . Extend these maps by the identity on boundary field coefficients.

To check all higher identities, let  $\mathcal{W}$  be the interval cochains with these field coefficients and let  $\mathcal{A}$  be their polynomial-form resolution. The coefficient algebra is the commutative algebra of represented zero-level current fields at the chosen separated points, interpreted coefficientwise in the insertion variables. Write  $sa$  for a suspension, with  $|sa| = |a| - 1$ . Put

$$D_1(sa) = s(da), \quad D_2(sa, sb) = (-1)^{|a|} s(ab).$$

Extend these maps to the direct sum of finite tensor words by inserting them into each consecutive block, with the Koszul sign from the preceding suspended inputs. This is a coderivation for the coproduct

which splits words into initial and final segments. Associativity and the Leibniz rule say  $(D_1 + D_2)^2 = 0$ . Use the same letters  $i, p, h$  for the suspended linear maps and define

$$F_1 = i, \quad R_n = \sum_{a+b=n} D_2(F_a \otimes F_b), \quad F_n = -hR_n, \quad b_n = pR_n \quad (n \geq 2),$$

with  $b_1 = sds^{-1}$ . An  $\mathcal{A}_\infty$  structure here means that the resulting degree-one coderivation  $b$  on the tensor words in  $sW$  satisfies  $b^2 = 0$ . The ordinary operations are fixed by

$$b_n(sa_1, \dots, sa_n) = (-1)^{\sum_j (n-j)|a_j|} sm_n(a_1, \dots, a_n).$$

Here is a direct verification of the recursion. The maps  $F_n$  extend uniquely to a coalgebra map  $F$ . Induct on word length, assuming  $DF = Fb$  and  $b^2 = 0$  in smaller lengths, where  $D = D_1 + D_2$ . With the length- $n$  terms temporarily zero, the length- $n$  morphism defect is

$$Q_n = R_n - L_n, \quad L_n = \sum_{\substack{r+q+t=n \\ 2 \leq q \leq n-1}} F_{r+1+t}(1^{\otimes r} \otimes b_q \otimes 1^{\otimes t}).$$

The tensor substitutions carry the insertion signs just specified. Let  $O_n$  be the length- $n$  term of  $b^2$  with  $b_n = 0$ . The identity

$$D(DF - Fb) + (DF - Fb)b = -Fb^2$$

gives  $\delta Q_n = -iO_n$ , where  $\delta Q_n = D_1 Q_n + Q_n b_1$  and  $b_1$  also denotes its extension to words. For  $k \geq 2$ ,  $pF_k = hF_k = 0$ , by  $ph = h^2 = 0$ . It follows that  $pL_n = hL_n = 0$ . Setting  $b_n = pQ_n = pR_n$  gives  $\delta b_n = -O_n$ , which is the missing length- $n$  identity for  $b^2$ . Setting  $F_n = -hQ_n = -hR_n$  and using  $D_1 h + hD_1 = 1 - ip$  gives

$$\delta F_n = -Q_n + ipQ_n + h\delta Q_n = ib_n - Q_n,$$

because  $hi = 0$ . This is the missing morphism identity. The induction starts with the cochain maps  $i, p$ , so it proves every arity without an infinite sum on any word. The linear map  $i$  is a quasi-isomorphism by the explicit contraction.

Every field comparison intertwines each multiplication vertex and each interval homotopy separately. It therefore intertwines these operations in every arity, before summing their binary trees. For example,

$$h(tdt) = \frac{t^2 - t}{2}, \quad \int_0^1 \frac{t - t^2}{2} dt = \frac{1}{12}$$

gives, in ordinary cohomological degrees  $|m_n| = 2 - n$ ,

$$m_3(e_1 J(z_1), \eta J(z_2), \eta J(z_3)) = -\frac{\eta}{12} J(z_1) J(z_2) J(z_3).$$

The degree-one output reflects the interval cochains. It is compatible with the strict degree-zero vertex algebra because this interval resolution has nonzero differential. It is not a cubic interaction in the BF action.

The quantum boundary map above identifies actual supported observables on the boundary with their bulk-boundary pushforward. The current comparison identifies its polynomial chiral vacuum, and the interval argument identifies the transferred separated products. These three maps have different sources. In particular, the last map alone does not describe a bulk action, a collision-chain compactification, or higher operations of an interacting theory. The zero-level BF construction supplies a physical realization of the zero-level current while retaining the additional bulk gauge and field cohomology computed above.

## Free fields on an interval

A free action determines Gaussian contractions. It does not determine a multiplication on a chosen complex of topological cochains. The distinction already appears when a field has values at two endpoints and a descendant integrated along the interval between them. Linear interpolation gives a binary product which is associative only up to a specified ternary operation. For a Heisenberg current, that operation retains the double pole of the operator product expansion. Its coefficient is an elementary integral.

The construction below keeps the holomorphic insertion points distinct. It gives all ordered higher operations on the polynomial de Rham model of an interval, with coefficients in the multilocal Wick algebra. An explicit comparison identifies the Wick factors with radially ordered free fields. Topological cochain degree, holomorphic jet order, and powers of the quantum parameter remain separate throughout.

### 2.1 THE CURRENT AND ITS SEPARATED PRODUCTS

Work on the affine complex curve with coordinate  $z$ . Put  $K = \mathbb{C}[\kappa][[\hbar]]$ , where both parameters have cohomological degree zero. The rank-one Heisenberg vacuum module over  $K$  is

$$V = \mathbb{C}[\kappa][q_1, q_2, \dots][[\hbar]].$$

Each coefficient of  $\hbar$  is a polynomial in finitely many  $q_m$ . All vacuum generators and holomorphic jet symbols have cohomological degree zero. For  $m > 0$ , define operators

$$J_{-m} = q_m, \quad J_m = \hbar \kappa m \frac{\partial}{\partial q_m}, \quad J_0 = 0.$$

The vacuum is  $\mathbf{1} = 1$ . Direct differentiation gives

$$[J_m, J_n] = \hbar \kappa m \delta_{m+n,0} \text{ id}.$$

The field  $J(z) = \sum_{n \in \mathbb{Z}} J_n z^{-n-1}$  splits as  $J_-(z) + J_+(z)$ , with creation modes in  $J_-$  and annihilation modes in  $J_+$ . In the formal expansion  $|z| > |w|$ ,

$$[J_+(z), J_-(w)] = \hbar \kappa \sum_{m \geq 1} m z^{-m-1} w^{m-1} = \frac{\hbar \kappa}{(z-w)^2}.$$

Here and below, a rational expression denotes its specified formal Laurent expansion. No norm convergence of unbounded operators is asserted.

Normal ordering places every creation mode to the left of every annihilation mode. The field attached to  $q_{r_1+1} \cdots q_{r_a+1} \mathbf{I}$  is

$$: \prod_{\nu=1}^a \frac{\partial_z^{r_\nu} J(z)}{r_\nu!} :.$$

The denominators identify mode indices with divided holomorphic jets. All holomorphic derivatives are retained.

For a finite set  $S$  of insertion labels, define

$$R_S = \mathbb{C}[z_i \ (i \in S), (z_i - z_j)^{-1} \ (i \neq j)], \quad W_S = (R_S[\kappa][x_{i,r} \mid i \in S, r \geq 0])[[\hbar]].$$

Tensor products of field modules are over  $K$ , completed in  $\hbar$ , with the displayed extension of rational scalars on the output. The symbol  $x_{i,r}$  represents the undivided derivative  $\partial_{z_i}^r J(z_i)$  inside one globally normal-ordered expression. Every coefficient is a polynomial in finitely many jet symbols. For disjoint sets  $S, T$ , extend scalars to  $R_{S \cup T}$  and set

$$P_{S,T} = \sum_{i \in S, j \in T} \sum_{r,s \geq 0} C_{ij}^{rs} \frac{\partial}{\partial x_{i,r}} \frac{\partial}{\partial x_{j,s}},$$

$$C_{ij}^{rs} = \hbar \kappa \partial_{z_i}^r \partial_{z_j}^s (z_i - z_j)^{-2}, \quad f \star g = \exp(P_{S,T})(fg).$$

The derivatives in  $P_{S,T}$  act only on jet symbols. They do not act on its rational coefficients. On polynomial inputs the exponential is finite. For formal inputs it is defined coefficientwise in  $\hbar$ .

Fix a total order  $i_1, \dots, i_a$  on  $S$ , with  $|z_{i_1}| > \cdots > |z_{i_a}|$  as a formal expansion convention. Writing  $V_o = \mathbb{C}[\kappa][q_1, q_2, \dots]$ , define

$$N_S : W_S \longrightarrow \text{Hom}_{\mathbb{C}[\kappa]}(V_o, V_o((z_{i_1})) \cdots ((z_{i_a})))[[\hbar]]$$

by replacing each jet monomial with its globally normal-ordered field and expanding each rational coefficient in that order. A fixed vacuum polynomial involves only finitely many annihilation modes, so the indicated iterated Laurent series are defined coefficientwise.

**Proposition 2.1** (Associativity and the field comparison). *For pairwise disjoint  $S, T, U$ , the two maps*

$$W_S \otimes W_T \otimes W_U \longrightarrow W_{S \cup T \cup U}, \quad (f \star g) \star h, \quad f \star (g \star h)$$

*are equal. In a radial ordering with every point of  $S$  outside every point of  $T$ , the normal-ordering map  $N$  satisfies*

$$N(f \star g) = N(f)N(g).$$

*Both identities hold for every holomorphic jet and every power of  $\hbar$ .*

*Proof.* All contraction operators commute: their coefficients are independent of the jet symbols being differentiated. Moreover,  $P_{S \cup T, U} = P_{S, U} + P_{T, U}$ . Each parenthesization is therefore

$$\exp(P_{S,T} + P_{S,U} + P_{T,U})(fgh).$$

This proves associativity without identifying distinct insertion points.

For the comparison, commute each annihilation mode in  $f$  past each creation mode in  $g$ . The displayed annihilation-creation commutator gives the contribution of one commutation. Its derivatives give  $C_{ij}^{rs}$ . Every commutator is central, so further commutations do not alter it. A collection of  $a$  disjoint contractions occurs  $a!$  times in  $P_{S,T}^a$ ; division by  $a!$  counts it once. Uncontracted modes remain in normal order. This proves the formula on monomials, hence on polynomials. Formal extension is coefficientwise: each input coefficient has finite polynomial degree, and only finitely many powers of  $\hbar$  contribute to a fixed coefficient.  $\square$

For each fixed radial order, put  $W_S^{\text{fld}} = \text{im } \mathcal{N}_S$ . The displayed comparison identifies its product with composition of the actual field operators on ordered blocks. If  $\mathcal{N}_S(f) = \mathbf{o}$ , then  $\mathcal{N}_{S \cup T}(f \star g) = \mathbf{o}$  for every following block  $T$ . The same holds for a zero factor in the second position. Thus the kernels are preserved by all ordered products. Every interval operation constructed below is a scalar cochain operation multiplied by such a product. It therefore descends to  $C_I \otimes W_S^{\text{fld}}$ , with the same contraction and proofs. This quotient supplies the operations on the actual free fields, including any relations among their normally ordered expressions.

Write  $x_i = x_{i,\mathbf{o}}$  and  $C_{ij} = \hbar\kappa/(z_i - z_j)^2$ . The first products are

$$\begin{aligned} x_1 \star x_2 &= x_1 x_2 + C_{12}, \\ x_1 \star x_2 \star x_3 &= x_1 x_2 x_3 + C_{12} x_3 + C_{13} x_2 + C_{23} x_1. \end{aligned}$$

Vacuum expectation means taking the scalar term after normal ordering. Two copies of  $J$  at the first insertion give

$$\langle : J^2 : (z_1) J(z_2) J(z_3) \rangle_{\text{conn}} = 2C_{12} C_{13}.$$

For three quadratic insertions the connected expectation is  $8C_{12} C_{13} C_{23}$ . There are two assignments of the first pair of fields to the other two vertices. At each remaining vertex there are two choices of the field used for its first edge. This gives  $2 \cdot 2 \cdot 2 = 8$  triangle pairings. Thus a Gaussian theory has nonzero connected composite-field products.

## 2.2 ENDPOINT VALUES AND AN INTEGRATED DESCENDANT

Give the interval  $I = [\mathbf{o}, \mathbf{1}]$  its increasing orientation. Its polynomial forms are the differential graded algebra

$$A_I = \mathbb{C}[t, dt], \quad |t| = \mathbf{o}, \quad |dt| = \mathbf{1}, \quad dt = dt, \quad (dt)^2 = \mathbf{o}.$$

The complex of cellular cochains is

$$C_I^{\mathbf{o}} = \mathbb{C}e_{\mathbf{o}} \oplus \mathbb{C}e_{\mathbf{1}}, \quad C_I^{\mathbf{1}} = \mathbb{C}\eta, \quad de_{\mathbf{o}} = -\eta, \quad de_{\mathbf{1}} = \eta, \quad d\eta = \mathbf{o}.$$

Here  $e_{\mathbf{o}}, e_{\mathbf{1}}$  record endpoint values and  $\eta$  records integration along the oriented edge. There are no endpoint vanishing conditions. Define inclusion and projection by

$$\begin{aligned} i(e_{\mathbf{o}}) &= \mathbf{1} - t, & i(e_{\mathbf{1}}) &= t, & i(\eta) &= dt, \\ p(f) &= f(\mathbf{o})e_{\mathbf{o}} + f(\mathbf{1})e_{\mathbf{1}}, \\ p(g \, dt) &= \left( \int_{\mathbf{o}}^{\mathbf{1}} g(t) \, dt \right) \eta. \end{aligned}$$

The degree-minus-one homotopy is

$$h(f) = \mathbf{o}, \quad h(g \, dt) = \int_{\mathbf{o}}^t g(u) \, du - t \int_{\mathbf{o}}^{\mathbf{1}} g(u) \, du.$$

It subtracts the linear function with the same endpoint difference.

**Lemma 2.2** (The interval contraction). *The maps above satisfy*

$$pi = \mathbf{1}, \quad dh + hd = \mathbf{1} - i \, p, \quad hi = ph = h^2 = \mathbf{o}.$$

*Proof.* The inclusion and projection commute with  $d$  by the fundamental theorem of calculus. On  $f(t)$ , the operator  $hd$  gives  $f(t) - f(o) - t(f(i) - f(o))$ . On  $g(t)dt$ , the operator  $dh$  gives  $(g(t) - \int_0^1 g)dt$ . These are exactly  $1 - i$  and  $p$  in their respective degrees. The function  $h(gdt)$  vanishes at both endpoints, proving  $ph = o$ . Also  $h(dt) = o$ ,  $h$  vanishes on functions, and its image consists of functions. These observations give  $hi = h^2 = o$ .  $\square$

Use the complexes

$$A(S) = A_I \otimes W_S, \quad C(S) = C_I \otimes W_S,$$

completed coefficientwise in  $\hbar$ . Their differential acts on the interval factor only. For disjoint supports their product on  $A$  is

$$(\alpha f)(\beta g) = (\alpha \wedge \beta)(f \star g).$$

Proposition 2.1 makes this an associative system of differential graded products. The maps  $i, p, h$  act on its interval factor and preserve each holomorphic support. These are topological forms with chiral field coefficients. This specified algebraic model does not identify the interval differential with a quantum Batalin–Vilkovisky differential.

The projected binary product  $m_2 = p\mu(i \otimes i)$  obeys

$$\begin{aligned} m_2(e_a, e_b) &= \delta_{ab}e_a, & m_2(e_a, \eta) &= m_2(\eta, e_a) = \tfrac{1}{2}\eta, \\ m_2(\eta, \eta) &= o & (a, b \in \{o, i\}). \end{aligned}$$

Every entry is multiplied by the corresponding Wick product of its field coefficients. In particular,

$$m_2(m_2(e_i, e_i), \eta) - m_2(e_i, m_2(e_i, \eta)) = \tfrac{1}{4}\eta.$$

Linear interpolation therefore forces a higher homotopy.

### 2.3 THE HIGHER OPERATIONS AND THEIR SIGNS

For a cochain  $a$ , write  $sa$  for its suspension, with  $|sa| = |a| - 1$ . A sequence of ordered operations is encoded by maps  $b_n$  of degree one on suspended inputs. For pairwise disjoint supports these satisfy

$$\sum_{r+q+t=n} (-1)^{|u_1|+\dots+|u_r|} b_{r+1+t}(u_1, \dots, u_r, b_q(u_{r+1}, \dots, u_{r+q}), u_{r+q+1}, \dots, u_n) = o.$$

All exponents use suspended degrees. The ordinary operations  $m_n$  of degree  $2 - n$  are defined by

$$b_n(sa_1, \dots, sa_n) = (-1)^{\sum_{j=1}^n (n-j)|a_j|} sm_n(a_1, \dots, a_n).$$

Here the tensor coalgebra has a precise support convention. Fix a finite ambient label set  $E$  and extend rational scalars to  $R_E$ . Take the direct sum of tensor words whose entries have pairwise disjoint supports  $S_1, \dots, S_n \subseteq E$ . The coproduct cuts a word into two consecutive pieces. Each operation replaces a consecutive block by one entry supported on its union. Thus all displayed compositions are defined over a common coefficient ring and remain in this coalgebra. The formulas commute with further extensions of  $E$ . An  $A_\infty$  structure here means a square-zero coderivation of these supported words. The words are finite, so no sum over arities is needed on an element.

On  $sA$ , put  $D_1(sa) = s(da)$  and  $D_2(sa, sb) = (-1)^{|a|}s(ab)$ . The differential graded algebra identities say  $D^2 = o$ , where  $D = D_1 + D_2$  is the induced coderivation. Write  $I, P, H$  for the suspended maps  $i, p, h$ . Starting with  $F_1 = I$ , define recursively, for  $n \geq 2$ ,

$$R_n = \sum_{a+b=n} D_2(F_a \otimes F_b), \quad F_n = -HR_n, \quad b_n = PR_n, \quad b_1 = sds^{-1}.$$

Each  $F_n$  has degree zero. Thus its tensor substitution contributes no extra sign. The only signs come from  $D_2$  and the coderivation insertion rule in the displayed higher identity.

**Theorem 2.3** (Ordered transfer with free-field coefficients). *The recursion defining  $R_n, F_n, b_n$  gives an uncurved  $A_\infty$  structure on the collection  $\{C(S)\}$ , with operations for every ordered list of pairwise disjoint supports. The maps  $F_n$  define an  $A_\infty$  quasi-isomorphism to  $\{A(S)\}$ . All identities preserve the rational holomorphic coefficients and their formal  $\hbar$  expansion. The element  $e_0 + e_1$  with empty field support is a strict unit.*

*Proof.* The proof takes place at each finite word length. Suppose the morphism equation  $DF = Fb$  and  $b^2 = 0$  hold in all smaller lengths. Set the length- $n$  components temporarily to zero. The length- $n$  morphism defect is

$$Q_n = R_n - L_n, \quad L_n = \sum_{\substack{r+q+t=n \\ 2 \leq q \leq n-1}} F_{r+1+t}(1^{\otimes r} \otimes b_q \otimes 1^{\otimes t}),$$

with the tensor signs fixed above. Let  $O_n$  be the length- $n$  component of  $b^2$  with  $b_n = 0$ . The identity

$$D(DF - Fb) + (DF - Fb)b = -Fb^2$$

gives  $\delta Q_n = -IO_n$ , where  $\delta Q_n = D_1 Q_n + Q_n b_1$  and  $b_1$  on the right denotes its extension to the input word. Lower defects vanish by induction. For  $k \geq 2$  the recursion gives  $PF_k = HF_k = 0$ . Consequently  $PL_n = HL_n = 0$ .

Set  $b_n = PQ_n = PR_n$ . Applying  $P$  to the defect identity gives  $\delta b_n = -O_n$ , which is exactly the length- $n$  equation  $b^2 = 0$ . Next set  $F_n = -HQ_n = -HR_n$ . The contraction identity and  $HI = 0$  give

$$\delta F_n = -Q_n + IPQ_n + H\delta Q_n = Ib_n - Q_n.$$

This is the missing length- $n$  morphism equation. Induction proves both assertions. The linear term  $I$  is a quasi-isomorphism by Lemma 2.2.

For the unit assertion, consider a binary transfer tree with a unit leaf. At its first vertex the unit removes that multiplication. If the other branch is a leaf, the next homotopy kills it by  $HI = 0$ . If that branch has an internal vertex, the next map gives  $H^2 = 0$  or  $PH = 0$ . Thus every operation of arity greater than two vanishes on a unit input. The binary and unary unit identities follow from the displayed table and  $d(e_0 + e_1) = 0$ .

Every tree has finitely many vertices. Associativity makes its field coefficient the same ordered Wick product, independently of the tree. The polynomial contraction acts over  $K$  and neither changes a holomorphic denominator nor a power of  $\hbar$ . The construction therefore extends to the stated coefficients and all holomorphic jets.  $\square$

The Wick exponential supplies every field coefficient of the transfer. The explicit interval contraction fixes the higher operations, including choices which a boundary identity alone cannot fix.

## 2.4 THE TERNARY OPERATION AND THE NEXT IDENTITIES

Write  $v = se_1$ ,  $w = s\eta$ , and  $\mathbf{e} = s(e_0 + e_1)$ . Their degrees are  $-1, 0, -1$ , respectively. The first suspended operations, with field coefficients suppressed, are

$$\begin{aligned} b_1(v) &= w, & b_1(w) &= 0, & b_2(v, v) &= v, & b_2(v, w) &= \frac{1}{2}w, & b_2(w, v) &= -\frac{1}{2}w, \\ b_3(v, w, w) &= \frac{1}{12}w, & b_3(w, v, w) &= -\frac{1}{6}w, & b_3(w, w, v) &= \frac{1}{12}w. \end{aligned}$$

Every other ternary value on  $v, w$  is zero. Values involving  $se_0$  follow from  $se_0 = \mathbf{e} - v$  and strict unitality.

The integral producing the first coefficient is

$$h(t \, dt) = \frac{t^2 - t}{2}, \quad F_2(v, w) = s \frac{t - t^2}{2}, \quad b_3(v, w, w) = s \left( \int_0^1 \frac{t - t^2}{2} \, dt \right) \eta = \frac{1}{12}w.$$

For the middle input there are two trees, each with the opposite sign. This gives  $-w/6$ . The last input gives the same value as the first.

**Example 2.4** (The Heisenberg ternary operation). For three distinct holomorphic insertions, ordinary unsuspension gives

$$\begin{aligned} m_3(e_1 x_1, \eta x_2, \eta x_3) \\ = -\frac{\eta}{12} \left( x_1 x_2 x_3 + \frac{\hbar \kappa x_3}{(z_1 - z_2)^2} + \frac{\hbar \kappa x_2}{(z_1 - z_3)^2} + \frac{\hbar \kappa x_1}{(z_2 - z_3)^2} \right). \end{aligned}$$

The output has topological degree one. The inputs have total degree two, so  $m_3$  has degree  $-1$  as required. The first term survives at  $\hbar = 0$ ; the other terms retain the quantum central contraction. Applying  $N$  identifies the parenthesis with  $J(z_1)J(z_2)J(z_3)$  in the chosen radial expansion.

The arity-three identity on  $(v, v, w)$  is an exact cancellation:

$$\underbrace{\frac{1}{2}w - \frac{1}{4}w}_{b_2 \text{ insertions}} + \underbrace{\left(-\frac{1}{6}w\right) - \frac{1}{12}w}_{b_1 \text{ insertions into } b_3} = 0.$$

It supplies the homotopy for the binary associator  $\eta/4$ . For  $(v, w, v, w)$  the arity-four identity is

$$-\frac{1}{12}w + \frac{1}{24}w - \frac{1}{24}w + \frac{1}{12}w = 0.$$

The first three summands insert  $b_2$  into  $b_3$ . The last inserts  $b_3$  into  $b_2$ . The remaining summand is zero. With field coefficients, every term multiplies the same fourfold Wick product by Proposition 2.1. These cancellations therefore verify the identities on the actual holomorphic expressions.

The quartic operation for this contraction vanishes identically. The following arity nevertheless requires a new operation. The next integrations give

$$\begin{aligned} F_3(v, w, w) &= s \left( \frac{t^3}{6} - \frac{t^2}{4} + \frac{t}{12} \right), \\ F_4(v, w, w, w) &= -s \frac{t^2(1-t)^2}{24}, \\ b_4(v, w, w, w) &= 0, \quad b_5(v, w, w, w, w) = -\frac{1}{720}w. \end{aligned}$$

Here  $\int_0^1 t^2(1-t)^2 dt = 1/30$ . On  $(v, w, w, v, w)$ , the three  $b_3$ -into- $b_3$  terms sum to

$$-\frac{1}{72}w - \frac{1}{144}w + \frac{1}{72}w = -\frac{1}{144}w.$$

The two nonzero insertions of  $b_1$  into  $b_5$  give  $w/180 + w/720 = w/144$ . Their coefficients follow from the five possible positions of the function input, computed below. Their cancellation proves the arity-five identity on an input where setting  $b_3 = 0$  would fail.

## 2.5 EVERY ARITY FROM ONE POLYNOMIAL INTEGRAL

Let the Bernoulli numbers be defined by

$$\frac{u}{e^u - 1} = \sum_{k \geq 0} B_k \frac{u^k}{k!}, \quad B_1 = -\frac{1}{2}.$$

For  $k \geq 1$ , put  $c_k = (-1)^k B_k / k!$ .

**Proposition 2.5** (Complete interval operations). *For  $k \geq 2$  and  $0 \leq j \leq k$ ,*

$$b_{k+1}(w^{\otimes j}, v, w^{\otimes(k-j)}) = (-1)^j \binom{k}{j} c_k w.$$

*All other operations of arity at least three on  $v, w$  vanish. These formulas, the binary table, and strict unitality determine every operation. For disjoint field supports each value is multiplied by the full ordered Wick product of its coefficients.*

*Proof.* Every nonzero higher transfer tree has only one function input. Indeed, every nonroot internal edge applies  $h$ , so its nonzero output is a function vanishing at the endpoints. A vertex cannot multiply two such outputs and then survive another  $h$ . Thus the tree must be a comb, with one leaf attached at each successive vertex. Every attached leaf before the root must be  $dt$ , since  $h$  kills functions. If the last leaf is a function, the root product is a function vanishing at both endpoints, which  $p$  kills. An all- $dt$  tree vanishes at its first vertex. This proves the asserted restriction on inputs.

Define  $T(q) = h(q dt)$  on polynomials. For the word  $(v, w, \dots, w)$  there is one comb, and its root coefficient is

$$c_k = \int_0^1 (-T)^{k-1} t dt.$$

To evaluate all these integrals, set  $G(u, t) = \sum_{a \geq 0} u^a (-T)^a t$  and  $M(u) = \int_0^1 G(u, t) dt$ . The equation  $G = t - uTG$  implies

$$\partial_t G = 1 - uG + uM, \quad G(u, 0) = 0, \quad G(u, 1) = 1.$$

Solving this formal differential equation gives

$$G(u, t) = \frac{1 - e^{-ut}}{1 - e^{-u}}, \quad M(u) = \frac{1}{1 - e^{-u}} - \frac{1}{u} = \sum_{k \geq 1} \frac{(-1)^k B_k}{k!} u^{k-1}.$$

The apparent poles at  $u = 0$  cancel, so these are formal power series. Finally, placing  $j$  copies of  $w$  before  $v$  gives  $\binom{k}{j}$  combs. Each left attachment contributes a minus sign from  $D_2(s dt, sf) = -s(f dt)$ . Right attachments contribute no sign. Their common integral is  $c_k$ , proving the formula.  $\square$

The scalar formulas follow from the interval integrals, while the field coefficients enter through the proved Wick comparison. At  $\kappa = 0$  or  $\hbar = 0$ , the contraction terms disappear while the topological higher products remain. Thus this higher arity is not a measurement of quantum loop order.

## 2.6 THE BETA-GAMMA FIELDS AND THE CURVE OPERATION

For the bosonic beta-gamma system use the base  $\mathbb{C}[[\hbar]]$  and replace the Heisenberg symbols by  $x_{i,r}^\beta, x_{i,r}^\gamma$ . Its elementary contractions are

$$C_{ij}^{\beta\gamma} = \frac{\hbar}{z_i - z_j}, \quad C_{ij}^{\gamma\beta} = -\frac{\hbar}{z_i - z_j}, \quad C_{ij}^{\beta\beta} = C_{ij}^{\gamma\gamma} = 0.$$

Jet contractions are their corresponding holomorphic derivatives. These coefficients are symmetric under simultaneous exchange of the point and field labels, as bosonic locality requires.

For completeness, take modes  $\beta(z) = \sum_m \beta_m z^{-m-1}$  and  $\gamma(z) = \sum_n \gamma_n z^{-n}$  with  $[\beta_m, \gamma_n] = \hbar \delta_{m+n,0}$ . The vacuum is killed by  $\beta_m$  for  $m \geq 0$  and by  $\gamma_n$  for  $n \geq 1$ . Explicitly, the vacuum module is  $\mathbb{C}[q_m (m \geq 1), p_n (n \geq 0)] [[\hbar]]$ , with

$$\beta_{-m} = q_m (m \geq 1), \quad \gamma_{-n} = p_n (n \geq 0), \quad \beta_m = \hbar \partial_{p_m} (m \geq 0), \quad \gamma_n = -\hbar \partial_{q_n} (n \geq 1).$$

Summing the two geometric series gives the displayed contractions. The proof of Proposition 2.1 therefore applies without change to the enlarged symbol set.

In particular, writing  $\beta_i, \gamma_i$  for its undifferentiated symbols, the ternary operation is

$$m_3(e_1\beta_1, \eta\gamma_2, \eta\beta_3) = -\frac{\eta}{12} \left( \beta_1\gamma_2\beta_3 + \frac{\hbar\beta_3}{z_1 - z_2} - \frac{\hbar\beta_1}{z_2 - z_3} \right).$$

The same scalar higher operations therefore act on both a double-pole current and a simple-pole first-order system, with their different holomorphic contractions intact.

There is a further operation when insertion points collide. In the Heisenberg vertex algebra, the regular coefficient is  $a_{(-1)}b = \text{Res}_{z=0} z^{-1}Y(a, z)b dz$ . This residue extracts a Laurent coefficient and has contour normalization  $1/(2\pi i)$ . For three currents, direct normal ordering and Taylor expansion give

$$((J_{(-1)}J)_{(-1)}J) - J_{(-1)}(J_{(-1)}J) = \hbar\kappa \partial^2 J.$$

Indeed,  $:JJ:(z)J(w)$  contains  $2\hbar\kappa J(z)/(z-w)^2$ , whose regular coefficient contributes  $\hbar\kappa \partial^2 J$ . The reversed iterated regular product has no such term. Thus this regular product cannot be the binary multiplication of a degree-zero ordinary minimal  $\mathcal{A}_\infty$  algebra. The nonzero result also exhibits the precise information lost by replacing the separated Wick product with coincident commutative symbols.

## 2.7 TOPOLOGICAL RESOLUTION AND BOUNDARY OBSERVABLES

The interval construction has a strict comparison to the original multilocal field system. The map

$$\mathcal{W}_S \longrightarrow C(S), \quad f \longmapsto (e_o + e_i)f$$

is a strict  $\mathcal{A}_\infty$  quasi-isomorphism. Its image is the constant cochain, the differential kills it, and every higher operation on it vanishes by strict unitality in the interval factor. For  $a = o, i$ , evaluation  $\epsilon_a : C(S) \rightarrow \mathcal{W}_S$  is also a strict quasi-isomorphism: higher outputs have edge degree, and evaluation kills them. Writing  $\iota(f) = (e_o + e_i)f$ , the composite  $\epsilon_a \iota$  is the identity. The other composite has the explicit chain homotopy

$$dk_a + k_a d = 1 - \iota \epsilon_a, \quad k_o(\eta) = e_i, \quad k_i(\eta) = -e_o, \quad k_a(e_b) = o.$$

For example, on  $e_o, e_i, \eta$ , the first identity with  $a = o$  reads  $-e_i, e_i, \eta$  on both sides. This explains how the same free fields admit a strict model and the explicit nonzero higher operations calculated above.

For a physical realization of the algebraic specialization  $\kappa = k \in \mathbb{C}$ , use the split holomorphic BF family of Chapter 1 with bulk parameter  $\ell = -k$ . Its fields on  $\mathbb{C} \times \mathbb{R}_{\geq o}$  have separate partial-form and holomorphic-one-form components  $(\alpha, \beta)$ ; write  $d' = \bar{\partial} + d_t$  for the partial de Rham differential. They carry a fixed nondegenerate cross pairing and differential

$$Q_\ell(\alpha, \beta) = (d' \alpha, d' \beta + \ell \partial \alpha).$$

The chiral boundary condition is  $\iota^* \alpha = o$ . The integrated pairing requires at least one compactly supported argument, and the integrated action is defined on compact fields; local densities are defined on unrestricted smooth fields. For  $k \neq o$ , the field change  $\alpha + \ell^{-1} \beta$  identifies this theory with abelian Chern–Simons theory whose symmetric coefficient pairing is  $\langle e, e \rangle = \ell = -k$ .

The boundary quantum observable complex has the positive unary operator and contraction

$$D = \bar{\partial} + \hbar \Delta_{\mu_\ell}, \quad \mu_\ell(u, v) = -\ell \int_{\mathbb{C}}^{\text{res}} u \wedge \partial v, \quad \int^{\text{res}} = \frac{1}{2\pi i} \int.$$

Choose a smooth radial compact cutoff  $\chi$  equal to one near the origin. Its current is fixed by  $q_{n+1} = [-\bar{\partial} \chi / z^{n+1}]$ ,  $n \geq o$ . For separated evaluation forms  $u_a, u_b$  and their representatives  $e_a, e_b$  on a common outer radial annulus, the compact chain calculation gives

$$[u_a u_b] = [e_a e_b] - \frac{\hbar \ell}{(a-b)^2} \mathbf{1} = [e_a e_b] + \frac{\hbar k}{(a-b)^2} \mathbf{1}.$$

Thus it is the extracted level  $-\ell = k$  that matches the algebraic parameter of this chapter. The Cauchy conjugation and moment calculation in Chapter 1 transport every separated product to its Wick formula and recover the complete mode action

$$J_{-m} = q_m, \quad J_m = \hbar k m \partial_{q_m} \quad (m > o), \quad J_o = o, \quad J(z)J(w) \sim \frac{\hbar k}{(z-w)^2}.$$

The compact collar map intertwines the actual quantum differentials and all separated products. Consequently the represented fields have precisely the normalization used by the interval transfer. The extraction first uses polynomial  $\hbar$  and the finite-weight vacuum space. Formal completion is applied afterwards to the extracted modes and Wick products, coefficient by coefficient; on arbitrary completed

states the field target is the inverse limit of the Laurent-series spaces modulo  $\hbar^M$ . This does not identify full disk cohomology with its finite-weight vacuum or interchange cohomology and completion of the full observable complex.

At  $k = 0$ , the split field pairing remains nondegenerate and the boundary current becomes commutative, with all polynomial jet fields retained. The field change to the nonzero de Rham presentation contains  $\ell^{-1}$  and has no zero specialization. In particular, the zero symmetric pairing on a single Chern–Simons coefficient space supplies no inverse BV pairing or propagator. The zero-level physical realization uses the two separate BF field components.

The compact quantum contraction of Theorem 1.8 supplies a chain map from the finite current part of this interval model to the full boundary observable complex. Fix its finite list of points and divided jets, and set  $\kappa = -k$  in that theorem. Its normal polynomial algebra  $A = \mathbb{C}[y_1, \dots, y_N][\hbar]$  represents the common-annulus currents. An undivided jet of order  $n_i$  is represented by  $n_i! y_i$ . Insertion groups remain at distinct holomorphic points.

**Proposition 2.6** (The supported interval chain map). *Let  $\iota_U : A \rightarrow \mathcal{B}_{-k}(U)$  and  $L_U : \mathcal{B}_{-k}(U) \rightarrow A$  be the compact inclusion and moment projection of Proposition 1.9. For  $a = 0, 1$ , define*

$$\Phi_a : C_I \otimes A \longrightarrow \mathcal{B}_{-k}(U), \quad \Phi_a(\alpha \otimes F) = \epsilon_a(\alpha) \iota_U(F), \quad \Psi(b) = (e_0 + e_1) \otimes L_U(b).$$

*These are chain maps for the interval differential and the actual quantum differential, respectively. They satisfy*

$$\Phi_a \Psi = \iota_U L_U, \quad 1 - \Psi \Phi_a = d(k_a \otimes 1) + (k_a \otimes 1) d \quad \text{on } C_I \otimes A.$$

*Their induced map identifies  $H^\bullet(C_I \otimes A) = A$  with the annular polynomial subspace of  $H^\circ(\mathcal{B}_{-k}(U))$ . For every tuple of polynomial currents at distinct points, the comparison of its supported product with the annular Wick expression has the compact primitive  $\rho H F$  in (1.11).*

*Proof.* Endpoint evaluation kills  $dC_I$  and  $\iota_U(F)$  is a quantum cycle. The map  $L_U$  intertwines the quantum differential with zero. These statements prove the chain-map equations. The equality  $L_U \iota_U = 1$  and the displayed interval contraction prove both composite identities. The split inclusion proves the cohomology assertion. The product statement is (1.11), with  $-\hbar\kappa = \hbar k$ .  $\square$

One can retain the interval factor during the compact contraction. On  $C_I \otimes Q$ , use the total differential  $d(\alpha \otimes q) = d_I \alpha \otimes q + (-1)^{|\alpha|} \alpha \otimes D_Q q$ . Then

$$\mathcal{H}(\alpha \otimes q) = (-1)^{|\alpha|} \alpha \otimes Hq, \quad d\mathcal{H} + \mathcal{H}d = 1 - (1 \otimes \iota_U).$$

The two terms containing  $d_I \alpha$  cancel by their opposite signs. The remaining terms are exactly the compact quantum contraction. Thus this extension keeps the interval differential distinct from the quantum differential while intertwining both.

The map  $\Phi_a$  is a map of complexes. The product comparison uses compact homotopies because common-annulus representatives overlap. In particular, the second-order quantum differential does not define an ordinary differential graded multiplication on that full target. An ordered higher morphism requires compatible Taylor maps on these same supported complexes. A collision comparison also requires operations on collision chains and their boundary strata. Neither construction follows from the linear chain map alone.

Ordered defect operations and symmetric perturbative brackets likewise require distinct insertion rules: the former retain the order of inputs along the defect, while the latter sum over permutations with

their graded signs. The Wick field comparison preserves ordered products at distinct points. Its supported realization has the compact chain homotopies above. A comparison with renormalized configuration integrals must construct compatible Taylor maps on the same complex and verify their equations at every arity. Neither the Gaussian contractions nor the interval identities alone supply the quantum master equation for an interacting theory.